PRESENTING

AARON

UC SAN DIEGO
AIAA DBF 25-26
DESIGN REPORT

Table of Contents

NOMENCLATURE **2**

1. EXECUTIVE SUMMARY **3**

2. MANAGEMENT SUMMARY **4**

2.1 Team Organization and Roles 4
2.2 Project Timeline and Major Milestones 5

3. CONCEPTUAL DESIGN **6**

3.1 Competition Overview and Scoring 6
3.2 Subsystem Requirements Identification 7
3.3 Score Sensitivity Analysis 8
3.4 Aircraft Configuration Selection 8
3.5 Initial Sizing Estimate 14

4. PRELIMINARY DESIGN **14**

4.1 Design Methodology 15
4.2 System-level Sizing Optimization 16
4.3 Aerodynamic Design 22
4.4 Propulsion System Selection (SSO $i = 4$) 30
4.5 Performance Predictions 32

5. DETAILED DESIGN **33**

5.1 Aircraft Dimensions 33
5.2 Structural Design and Analysis 33
5.3 Avionics and Wiring 38
5.4 Weight and Balance 39
5.5 Drawing Package 39

6 MANUFACTURING PLAN **44**

6.1 Selection of Manufacturing Processes 44
6.2 Material Selection 45
6.3 Manufacturing Timeline 47

7. TESTING PLAN **48**

7.1 Testing Schedule 49
7.2 Sub-system Testing 49
7.3 Flight Testing Plan 51
7.4 Risk Reduction and Preflight Checklist 53

8. PERFORMANCE RESULTS **54**

8.1 Subsystem Performance 54

9. BIBLIOGRAPHY **56**

Nomenclature

Important Acronyms and Mathematical Symbols

UC	University of California
DBF	Design/Build/Fly
AIAA	American Institute of Aeronautics and Astronautics
AARON	Another Airplane Running On Nothing
GM	Ground mission
M1, 2, 3	Missions 1, 2, and 3
AVL	Athena Vortex Lattice
OpenVSP	Open Vertical Sketch Pad
MATLAB	Matrix Laboratory
CG	Center of gravity
Eq.	Equation
FOM	Figure(s) of merit
COTS	Commercial off-the-shelf
AoA	Angle of attack
MDO	Multidisciplinary design optimization
OML	Outer mold line
TVC	Tail volume coefficient
LiPo	Lithium-ion polymer
RC	Radio/Remote-control
APC	Advanced Precision Composites

S_C	Competition score
S_P	Participation score
TMS	Total mission score
S_{TR}	Total report score
S_{PR}	Proposal score
S_{DR}	Design report score
S_{GM}	Ground mission score
$S_{M1/M2/M3}$	Mission 1/2/3 score
S_P	Total participation score
$S_{P,A/TI/FM}$	Participation scores for attendance/ technical inspection/ flight mission
$t_{\min}$	Minimum recorded ground mission time
t_{UCSD}	Team ground mission time
N_P, N_C, N_L	Number of cargo/ passengers/ laps flown
EF	Battery efficiency factor
RAC	Rated aircraft cost
b	Wingspan
L_{Bann}	Banner length
E_{Batt}	Battery energy capacity
AR	Aspect ratio
W_o	Gross weight
S	Wing planform area
L/D	Lift-to-drag ratio
$C_{\ell/d/m}$	2D lift/drag/moment coefficients
$C_{L/D/M}$	3D lift/drag/moment coefficients
Re	Reynolds number
α	Angle of attack
λ	Taper ratio
f_b	Inboard wing section span fraction
L_F	Fuselage length
x_W	Wing leading edge position
$S_{H/V}$	Horizontal/vertical stabilizer planform area
SM	Static Margin
$x_{CP/AC/NP}$	Position of center of pressure/ aerodynamic pressure/ neutral point
FR	Fineness ratio
FF	Form factor
t/c	Thickness-to-chord ratio
$\omega_{n/d}$	Natural/damped frequency
σ	Time constant

1. Executive Summary

The University of California (UC), San Diego Design/Build/Fly (DBF) team presents its preliminary aircraft concept in this design report for the American Institute of Aeronautics and Astronautics (AIAA) 2026 DBF competition, outlining the sizing methodology, aerodynamic/propulsive optimization approach, structural design strategy, fabrication plan, and mission-driven performance targets. The aircraft, named *AARON*, is engineered to satisfy the four competition tasks: a proof of flight mission (M1), a transport mission (M2) that rewards the number of passengers/cargo carried and laps flown within a limited flight window of 5 minutes, a banner towing mission (M3) that rewards the length of a banner featuring the university logo and the number of laps flown within the same window, and a ground mission (GM) that rewards the speed of payload installation and removal. The design restrictions, scoring formulae, and diverse payload requirements stipulated this year by AIAA motivate a careful compromise between several conflicting design objectives as well as an intelligent prediction of their correlation with a high total mission score at the competition site. Based on a sensitivity analysis that integrates statistical reduction of score distributions from similar prior DBF competitions, the team identified passenger capacity and cruise efficiency as objectives that maximize the total score with the highest certainty. A rigorous multidisciplinary optimization (MDO) process that integrates expert pilot opinion was used to converge on an aircraft with a maximum gross weight of approximately 15.5 lb and a maximum cruise speed of approximately 94 ft/s.

Using a logical sequence of trade studies, the team selected a conventional aircraft configuration with a fuselage wrapped around slender passenger and an upper- and lower-level cargo compartment. A single tractor motor is utilized to maximize cruising efficiency in both M2 and M3. The tricycle landing gear configuration is used to establish a low incidence angle to maximize cargo access points, and a low-wing design is used to maximize passenger access. Passengers will be secured with foam strips and a 3D-printed twin dome structure that accommodate their diverse geometries. The banner will be cut from lightweight woven nylon and stowed longitudinally beneath the fuselage to reduce parasitic drag during cruise and climb-out, respectively. Deployment and release mechanisms will employ simple 3-bar pin linkages to minimize weight and maximize reliability amidst strong Kansas gusts. To balance structural efficiency and survivability, the airframe is constructed using a strategic combination of materials and techniques: the fuselage and wing will be assembled out of balsa/basswood internals, carbon fiber spars, 3D-printed lightweight polylactic acid control surfaces, and covering film, and slender tail surfaces will be constructed by wrapping expanded polystyrene with carbon fiber reinforcement. Top-level aircraft specifications and mission-specific performance predictions are enumerated in Table 1.a.

Table 1.a: Specifications and Performance Predictions of the UC San Diego DBF 2026 Aircraft

Properties	M1	M2		M3	GM
Payload(s)	None	38 passengers, 12 cargo units		30.5-ft-long banner	–
Payload Weight (lb)	0	6.16		1.23	–
Gross Weight (lb)	9.05	15.05		10.23	–
Cruise Speed (ft/s)	65	94.5		36.3	–
Number of Laps	3	8		4	–
Mission Duration (sec)	131.8	252.2		268.0	133
Net Charter Income	1300.4	Efficiency Factor	0.999	Rated Aircraft Cost	1.0

2. Management Summary

DBF at UC San Diego is a student-run project team consisting of about 50 active members, three of whom are graduate students, five are undergraduate senior members, and the rest are new recruits. The team is financially supported by the UC San Diego Jacobs School of Engineering and by various sponsoring companies. Likewise, alumni, faculty members, and company sponsors advise the team at periodical design reviews.

2.1 Team Organization and Roles

The organizational structure of the team is outlined in Figure 2.1. Faculty and alumni offer guidance through monthly design reviews. The project manager is responsible for managing the team's communications and finances, with the aid of four leads in the logistics sub-team, which handles administrative tasks including rules compliance, finance, media, and documentation. The chief engineer oversees design optimization, manufacturing processes, and system integration, and manages the progress of the four technical sub-teams.

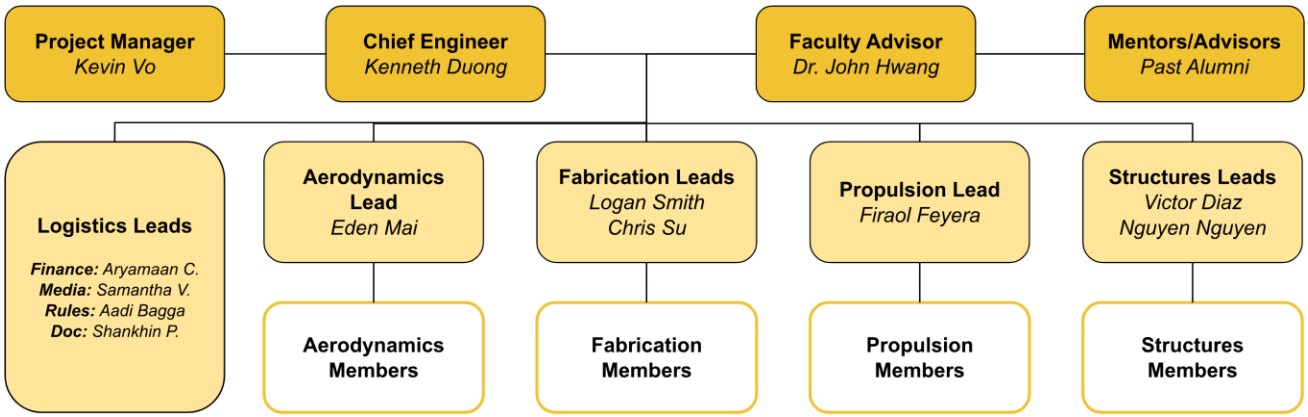


Figure 2.1: DBF at UC San Diego 2025-2026 Team Structure

Each technical sub-team is led by a student who hosts weekly meetings where they distribute project tasks among recruits and provide training to ensure the continued transfer of knowledge and succession for future leadership. The specific responsibilities and contributions of each sub-team are detailed in Table 2.a. Leads report progress to the project manager and chief engineer during weekly 'work-hour' meetings, which serve as the primary forum for interdisciplinary collaboration and finalization of design decisions.

Table 2.a: Sub-team Responsibilities Allocation

Sub-team	Responsibilities	Skillsets
Aerodynamics	Optimize the outer mold line through aerodynamics analyses of lift, drag, and stability, for each mission.	• Lift and stability analyses using XFLR5/AVL. • Mission-specific drag estimation using OpenVSP.
Propulsion	Select the optimal propeller(s) and motor using database searches, bench tests, and flight telemetry analysis.	• Power system selection using eCalc/MotoCalc. • Bench/flight test data analysis using MATLAB.
Structures	Analyze load cases, loft the detailed outer mold line, design the structural layout, and produce a CAD package for fabrication.	• Load path analysis and structural optimization. • Component design/assembly using SolidWorks. • Finite element analysis for component optimization.

Fabrication	Construct and test prototypes of critical mechanical subsystems and assemble the competition aircraft.	• Subsystem-level model-scale rapid prototyping. • Hand/power tools and specialized equipment. • Flight performance analysis and risk analysis.
Logistics	Support the Project Manager, advisor, and mentors in administrative tasks essential to the team's function.	• **Finance:** Budget balancing and sponsorship application. • **Media:** Promotional content and website maintenance. • **Rules Compliance:** Ensures deliverables satisfy rules/ Q&As. • **Documentation:** Formatting of proposal and design report.

2.2 Project Timeline and Major Milestones

The team's projected schedule for the 2025–2026 academic year is visualized in Figure 2.2 using a living Gantt chart, which represents the workflows of the five sub-teams in the context of major milestones. The team uses the Gantt Chart to monitor and track progress during the weekly leads' meetings, where any tasks which are falling behind schedule and significantly hindering team progress are identified and escalated to senior management. During monthly design reviews, the project manager and faculty advisor adjust the schedule as needed and redistribute task responsibilities to efficiently meet the final report and fly-off deadlines. As visualized in the figure, the project timeline defines nine key milestones that embody the team's primary goals, which are shown as red markers. Planned time frames allocated for each phase of the design, fabrication, and testing processes are indicated using purple bars. Yellow bars reflect the team's recorded progress at the time of submission of this report. Finally, a black marker at the end of a yellow bar signifies a completed task.

The Gantt chart calls for the manufacturing of a subsystem-level ground mockup as well as two separate aircraft: an aerodynamic prototype and the final competition plane. The objective of the 'Iron Bird' mockup is to reduce technical risks pertaining to the proposed mechanisms for installment, securing and removal of all payloads. The aerodynamic prototype serves as a low-cost platform for the team to assess the cruising efficiency, stability, and handling of the proposed aircraft under the various mission configurations. Pilot feedback and flight telemetry data will be used to refine the propulsion system, tail size, center-of-gravity (CG) location, and payload placement to achieve the desired performance for the final competition aircraft.

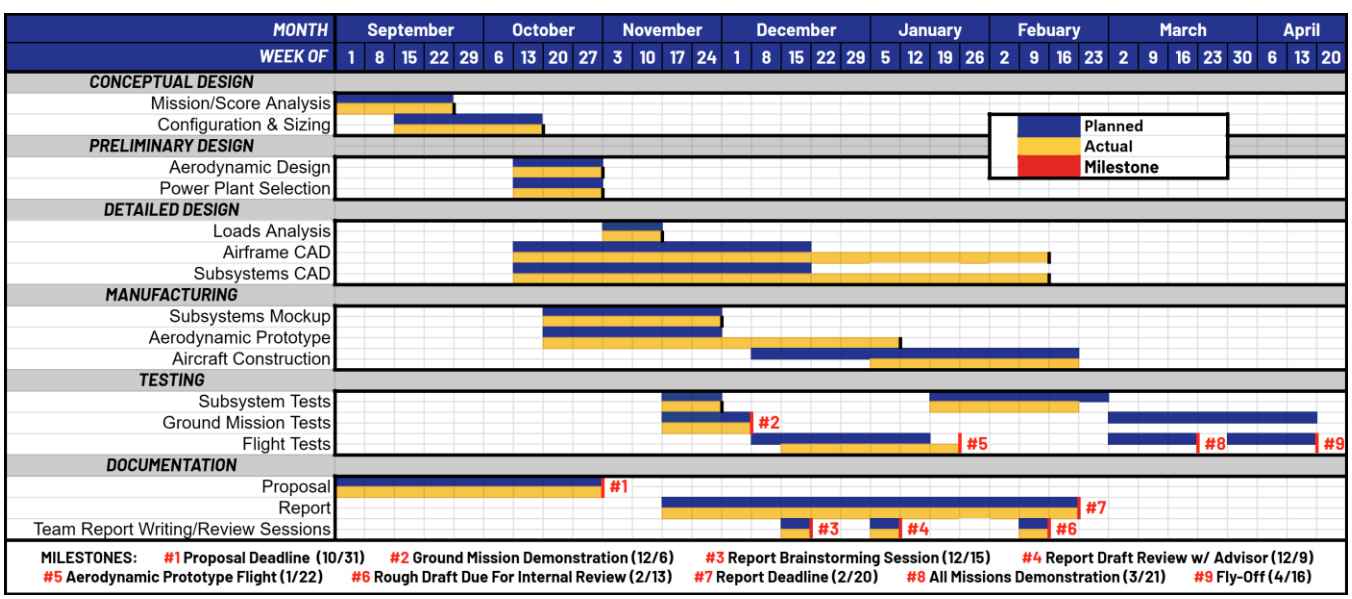


Figure 2.2: UCSD DBF 2025-2026 Gantt Chart

3. Conceptual Design

This section presents the problem statement of the 2025-2026 AIAA DBF competition [1] and outlines the strategic solution approach employed by the UC San Diego team. Mission requirements and design constraints were first identified, and a quantitative score sensitivity analysis was performed. Afterwards, subsystem-level design objectives were formulated to form the top-level basis of an initial aircraft design.

3.1 Competition Overview and Scoring

There are three flight missions (M1/M2/M3) and one GM in the competition. As shown in Eq. 3.1 (Equation 3.1), the competition score S_C is a function of a participation score S_P, the total mission score TMS, and the total report score S_{TR} which is based on the quality of the proposal (PR) and the design report (DR).

$$S_C = S_{TR} \times TMS + S_P \text{ , where } S_{TR} = 0.15S_{PR} + 0.85S_{DR} \text{ and } TMS = S_{GM} + S_{M1} + S_{M2} + S_{M3} \tag{3.1}$$

As shown in Eq. 3.2, one participation point will be given for each of the following: attending the fly-off (A), completing tech inspection (TI), and attempting a flight mission (FM). A maximum S_P of 3 may be earned.

$$S_P = S_{P.A} + S_{P.TI} + S_{P.FM} \tag{3.2}$$

The score for each mission is determined by several performance metrics, as detailed below.

3.1.1 Ground Mission

In the GM, the assembly crew members and pilot will be timed as they transition the empty aircraft into its flight-ready configuration and demonstrate remote banner deployment/release. The transition involves loading and unloading all declared passengers and cargo and installing the banner mechanism. The mission score S_{GM} will be determined using Eq. 3.3, where $t_{\min}$ represents the fastest mission time recorded among all teams.

$$S_{GM} = \frac{t_{\min}}{t_{\text{UCSD}}} \tag{3.3}$$

3.1.2 Flight Overview

The aircraft must follow the same generic flight path for every flight mission. The path consists of a 2,000-foot cyclical traffic pattern that includes an outwards 360-degree turn on the downwind segment. Figure 3.1 illustrates the pattern divided into distinct phases. A successful landing is required to receive a score in each flight mission.

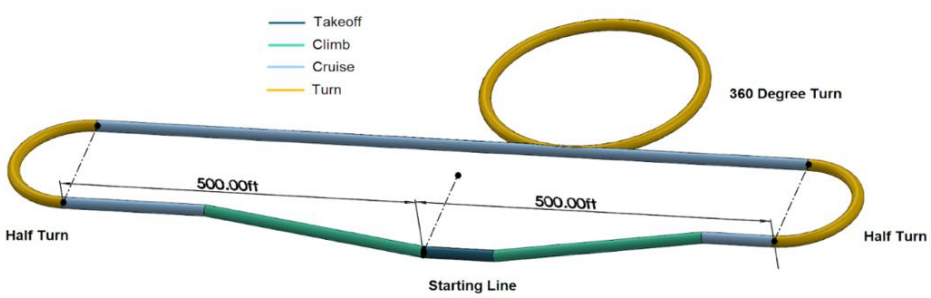


Figure 3.1: 3D Visualization of the AIAA DBF Competition Flight Path

3.1.3 Mission 1 – Test Flight

M1 verifies the aircraft's basic flight performance by requiring completion of three laps within the allotted 5-minute window. If completed, the mission earns the team one point as shown in Eq. 3.4.

$$S_{M1} = 1.0 \tag{3.4}$$

3.1.4 Mission 2 – Charter Flight

M2 requires the aircraft to complete as many laps as possible within a five-minute window while carrying as many passengers and cargo units as possible. The mission score awards the number of laps completed and the amount of payload transported but penalizes a high battery energy capacity, as described in Eq. 3.5–3.7.

$$S_{M2} = 1 + \frac{[\text{Net Income}]_{\text{UCSD}}}{[\text{Net Income}]_{\max}} \tag{3.5}$$

$$\text{Net Income} = N_P[6 + N_L(2 - 0.5EF)] + N_C[10 + N_L(8 - 2EF)] - 10N_L EF \tag{3.6}$$

$$EF = \text{Efficiency Factor} = \text{Total Propulsion Battery Capacity (Wh)} \;/\; 100 \tag{3.7}$$

3.1.5 Mission 3 – Banner Flight

M3 requires the aircraft to deploy and tow a banner, complete as many laps as possible in under five minutes and release the banner on the final lap. The mission score awards the number of laps flown and the banner length, but penalizes a large wingspan b through the rated aircraft cost RAC, as defined in Eq. 3.8 and 3.9.

$$S_{M3} = 2 + \frac{[N_L L_{\text{Bann}}/\text{RAC}\,]_{\text{UCSD}}}{[N_L L_{\text{Bann}}/\text{RAC}]_{\max}} \tag{3.8}$$

$$\text{RAC} = 0.75 + (0.05\text{ft}^{-1})b \tag{3.9}$$

3.2 Subsystem Requirements Identification

A comprehensive analysis of the mission requirements, summarized in Table 3.a, revealed potentially competing subsystem-level objectives that influenced the team's design strategy: namely, payload carrying capacity (M2) and payload installation/removal speed (GM), and aerodynamic/propulsive efficiency for two airspeeds (M2, M3) that are likely to differ greatly due to the towing of a competitively long banner in the latter [2].

Table 3.a: Mission- and Subsystem-Level Requirements

Mission	Competition Requirements	Deduced Subsystem Requirement	Req. #
M1) Test	• Prepare aircraft in a 5-min. staging period. • Complete laps in a 5-min. flight window. • Max wingspan & weight: 5 ft & 54.9 lb.	Hatches/switches are easily accessible.	*RQ-M1.1*
		Aircraft must be sufficiently strong, fast, stable and reliable to fly 3 laps in 5 min.	*RQ-M1.2*
M2) Charter	• Load propulsion battery & payload in 5 min. • Store at least 3 passengers per cargo unit. • Cargo may not be stored atop passengers. • Passengers will sit upright on a single plane. • Fly as many laps as possible in 5 min. • Land without bouncing off the runway (applies to all flight missions).	Payload storage systems must be light and compact to reduce weight and drag.	*RQ-M2.1*
		Wing/fuselage must be strong enough to support failure loads at max gross weight.	*RQ-M2.2*
		Propulsion system must be optimized to balance static thrust and cruise efficiency.	*RQ-M2.3*
		High static thrust and airframe tensile strength required to support banner drag.	*RQ-M2.4*

M3) Banner	• Load propulsion battery and stow banner (length > 10" & aspect ratio < 5) in 5 min. • Fly as many laps as possible in 5 min while towing banner in vertical orientation. • Deploy and release banner remotely. • Banner must remain intact after impact.	Banner material must be optimized for low drag, low weight and tear resistance.	*RQ-M3.1*
		Sufficient ballast is required to keep the banner bottom-heavy in flight.	*RQ-M3.2*
GM) Ground Mission	• Flight controls must remain operational. • All wheels must remain on the ground. • Load/unload passengers and cargo. • Stow the banner externally. • Suspend aircraft. Deploy/release the banner. • Complete tasks as quickly as possible.	All payload systems must be designed for reliable and fast loading/unloading.	*RQ-GM.1*
		Payload doors must allow easy access without compromising structural integrity.	*RQ-GM.2*
		Banner assembly must deploy/release rapidly under its own partial weight.	*RQ-GM.3*

3.3 Score Sensitivity Analysis

A detailed sensitivity analysis was performed to examine the relative influence of all critical scoring variables on the total mission score. As depicted in Figure 3.2, the M2 score was found to rise faster with increasing cargo count than lap count N_L, which is conducive to prioritizing payload weight over cruise speed due to the associated growth rates of induced drag energy. Thus, increasing the number of low-density passengers, N_P, was expected to induce a negligible drag penalty due to the growth rate of the sub-dominant parasitic drag with fuselage volume (two-thirds power). The M3 score rewards banner length L_{Bann} and speed similarly. However, prioritizing the former was preferred due to the lower growth rate of required power, up to a critical point beyond which induced drag due to high banner/ballast weight would reduce the score (*RQ-M3.2*). The scoring functions thus motivate (i) maximizing payload amount while compromising loading/unloading times, (ii) maximizing wing aspect ratio and energy capacity E_{Batt}, which improves range through aerodynamic efficiency and endurance, and (iii) maximizing wing loading, which reduces the maximum power requirement. Each of the tradeoffs was investigated with higher quantitative precision using a reduced-order flight mission model, which is detailed later in §4.1.

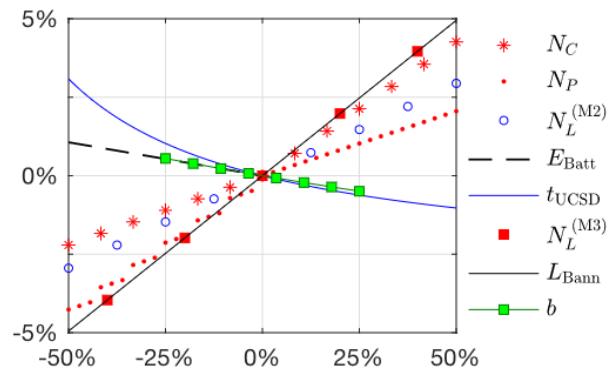


Figure 3.2: Score sensitivity analysis for TMS

3.4 Aircraft Configuration Selection

Based on the score and requirements analyses, the team identified eight key design decisions that define the aircraft's configuration as it pertains to competitive performance [3]. The decisions are outlined in Table 3.b.

Table 3.b: Configuration Trade Studies and Leads in Charge

No.	Trade Study	Configurations Explored	Leads in Charge
3.4.1	Aircraft Type	Monoplane, Blended Body, Twin Boom	Aerodynamics / Fabrication
3.4.2	Internal Payload Arrangement	Layers, Sequence, Hybrid	Fabrication / Structures
3.4.3	Fuselage Cross-section Shape	Rectangle, Rounded Rectangle, Tapered Down	Structures / Fabrication
3.4.4	Stowed Banner Placement	Above, Below, and Alongside Fuselage	Fabrication / Aerodynamics

3.4.5	Wing Placement	High, Mid, Low	Structures / Aerodynamics
3.4.6	Propulsion Configuration	Single Tractor, Dual Tractor, Single Pusher	Propulsion / Structures
3.4.7	Landing Gear	Tail Dragger, Tricycle, Retractable	Structures / Fabrication
3.4.8	Empennage	Standard, H-tail, T-tail, V-tail	Aerodynamics / Fabrication

The configuration decisions were organized into a logical sequence of trade studies wherein suitable designs were comparatively evaluated using figures of merit (FOM) deduced from mission/subsystem requirements. To ensure a robust selection free of personal bias, each decision was first investigated by the leads of the most pertinent sub-teams, as indicated in Table 3.a, after which the leads in charge reported to the chief engineer, pilot, faculty advisor, and alumni mentors for discussion and re-evaluation. Details regarding the decision process and FOM are provided below in Tables 3.c and 3.d, respectively.

Table 3.c: Configuration Selection Process

No.	Step	Description
1	**FOM Identification**	FOM in line with the primary subsystem-level design objectives are identified.
2	**Weighting**	FOM are weighted on a scale of 1:5 based on significance to mission performance.
3	**Configuration Identification**	Potentially advantageous configurations are scrutinized at a high level based on scholarly research and previous team experience.
4	**Scoring**	Configurations are scored on a scale of 1 to 5 (worst to best performance) based on their relative likelihood of success with regard to each FOM.
5	**Total Score**	Each option is evaluated based on the total score, which is the sum of the scores for the characteristics multiplied by their respective weights. The final configuration selected will have the highest total score. The total score formula is shown below: $\text{Total Score} = (\text{Weight \#1})\times(\text{Score \#1}) + (\text{Weight \#2})\times(\text{Score \#2}) + \ldots$

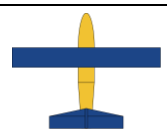
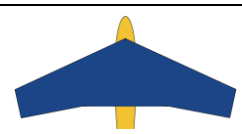
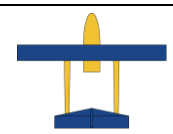
Table 3.d: Figures of Merit

Figure of Merit	Relevance to Mission Performance
Historical Record	Frequency of success in prior DBF competitions with similar mission requirements.
Manufacturability	Low complexity; easy to learn; time- and resource-efficient. Allows for an increased number of prototype iterations and ground/flight tests for improved final performance.
Internal Space	Conducive to compliant placement of cargo and (high-volume) passenger compartments.
Structure/Weight	Low structural weight and complexity required to ensure the strength and safety of the airframe in flight. Lower empty weight fraction enables greater payload amount in M2/M3.
Aerodynamics	Enable cruising efficiency and top speed conducive to a high number of laps in M2/M3.
Stability/Control	Handling characteristics allow competitive maneuvering with low/moderate pilot effort.
Serviceability	Quick, easy and reliable access to passenger/cargo compartments for a low GM time.
Tow Interference	Lower risk of interference between tow wire and airframe during deployment/maneuvers.
Takeoff	Ability of the aircraft to execute a takeoff rotation without the banner striking the runway.

The decision process is visualized below with use of Pugh Charts, where each FOM is assigned a weight from 1 (least important) to 5 (most important) based on its relevance to the specific decision under consideration.

3.4.1 Aircraft Type

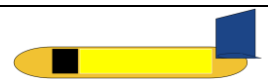
Table 3.e: Pugh Chart for Aircraft Type

Category	Weight	 Monoplane	 Blended Wing Body	 Twin Boom
Structure/Weight	**5**	5	4	3
Aerodynamics	**5**	4	5	3
Serviceability	**4**	5	4	4
Manufacturability	**4**	5	3	4
Historical Record	**3**	5	0	1
Stability/Control	**3**	5	3	5
Total	**/115**	**115**	**82**	**80**

The choice of airframe configuration was primarily driven by the high sensitivity of M2 and M3 to payload volume and banner length, which placed a strong emphasis on structural efficiency, internal volume, and ease of payload integration. When compared with a monoplane, a blended wing body may offer superior aerodynamic efficiency and internal volume if carefully optimized, but introduces significant challenges in structural design, manufacturing, and rules-compliant payload access. Similarly, the twin-boom configuration offers superior pitch stability and reduced trim drag due to the vertical stabilizers acting as endplates, but typically suffers from poorer structural efficiency, net lift-to-drag ratio and manufacturability. The conventional monoplane provides a well-balanced solution across all mission-relevant evaluation categories, offering straightforward manufacturing, predictable stability/control, and far superior payload access conformant to the competition rules. Although the monoplane configuration may not offer the highest possible aerodynamic efficiency, it was ultimately selected due to its simplicity, robustness, mission suitability, and–most crucially–an overwhelming record of proven success in prior DBF competitions with similar requirements.

3.4.2 Internal Payload Arrangement

Table 3.f: Pugh Chart for Internal Payload Arrangement

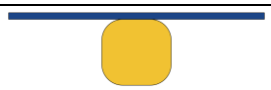
Category	Weight	Layers	 Sequence	Hybrid
Serviceability	**5**	4	4	3
Structure/Weight	**4**	5	3	4
Aerodynamics	**4**	3	5	4
Manufacturability	**4**	5	5	3
Control	**3**	5	2	3
Total	**/100**	**87**	**78**	**68**

As displayed in Table 3.f, the team explored various layouts for placement of internal payload (passengers colored yellow and cargo black). A layered arrangement consisting of a slender passenger compartment above a single, compact cargo hold was ultimately chosen to limit the physical movement required for the assembly crew and thus minimize t_{UCSD}. Unlike sequenced or hybrid arrangements with possibly more than one cargo hold, the selected arrangement also concentrates payload mass near the aircraft CG, thus preserving pitch authority, alleviating the fuselage bending moment, and improving payload capacity (S_{M2}) without increasing the required

structural weight. Although cargo access posed a serious challenge in terms of avoiding a hatch door that interrupts the main structural load path for fuselage bending, the layered configuration was deemed the most viable and competitive using the wholistic scoring process.

3.4.3 Fuselage Cross-section Shape

Table 3.g: Pugh Chart for Fuselage Cross-section Shape

Category	Weight	Rectangle	 Rounded Rectangle	Tapered Down
Manufacturability	**5**	5	4	3
Internal Space	**5**	5	4	3
Structure/Weight	**4**	3	5	4
Aerodynamics	**3**	3	4	5
Propulsion	**3**	3	4	5
Total	**/100**	**80**	**83**	**77**

A rounded rectangular fuselage cross section was selected, prioritizing payload volume and manufacturability without sacrificing aerodynamic and structural efficiency. This geometry maintains simplicity of manufacturing while limiting wetted area for a given internal volume, a critical factor for the high-volume passenger units required for the 2026 competition. The sharp corners of the pure rectangle cross section are favorable from a manufacturing standpoint but introduce undue stress concentration and interference drag. Unlike the tapered cross section, which is the most space-efficient due to the low volume of the cargo units and the limitation on cargo amount, the selected shape also simplifies integration of the banner mechanism below the fuselage.

3.4.4 Stowed Banner Placement

Table 3.h: Pugh Chart for Stowed Banner Placement

Category	Weight	Above Fuselage	Below Fuselage	Alongside Fuselage
Serviceability	**5**	5	3	4
Tow Interference	**4**	1	5	2
Takeoff	**3**	3	2	4
Aerodynamics	**2**	4	4	4
Total	**/70**	**46**	**49**	**48**

To ensure a successful banner deployment while minimizing aerodynamic penalties and maximizing assembly crew access, the team evaluated three stowage methods. A longitudinal orientation was enforced to accommodate a competitive banner length while maintaining a low-drag profile during the climb-out phase. The team ruled out mounting the banner alongside or above the fuselage due to the severe practical risk of airframe damage or failed deployment. In particular, a tail-strike by the ballast or snagging of the tow wire on moving ailerons may result in a total loss. On the other hand, placement below the fuselage offers the highest reliability of deployment, enables superior roll stability during climb-out, and allows stowage of longer banners without tail control interference or CG displacement due to heavy ballast. The primary drawbacks are operational: ground access is less convenient, and the pilot must execute a shallower takeoff rotation during M3 to prevent the banner from striking the runway and forcing mission forfeiture. Placing the burden of responsibility on pilot/assembly-crew training, the team deemed this configuration to be the most robust.

3.4.5 Wing Placement

Table 3.i: Pugh Chart for Wing Location

Category	Weight	High	Mid	Low
Serviceability	**5**	3	2	5
Structure/Weight	**4**	4	3	5
Aerodynamics	**4**	5	4	3
Manufacturability	**4**	5	3	4
Control	**3**	3	5	5
Total	**/100**	**80**	**65**	**88**

As reflected in Table 3.i, wing placement options were evaluated primarily based on serviceability and global structural efficiency. The high-wing and low-wing layouts are comparable in manufacturability, and so were preferred over a mid-wing configuration. The mid-wing was disqualified early in the selection process due to the high structural weight penalty incurred for a strong wing–fuselage joint and the loss of substantial internal space for a compact, contiguous cargo compartment. The significantly improved roll stability offered by a high-wing layout was considered of low relevance due to the nature of the flight course, which involves straight segments lasting only a few seconds amidst potentially severe gusts that may render passive stability moot. A low-wing design was ultimately selected to maximize passenger accessibility and reduce the structural weight associated with integration of a stable, wide landing gear base. Although this configuration results in reduced pendulum stability and an increased stall speed relative to a high-wing layout, these characteristics were deemed manageable through focused pilot training and careful design of the wing planform. Both assumptions were validated by way of early prototype testing.

3.4.6 Propulsion Configuration

Table 3.j: Pugh Chart for Propulsion Configuration

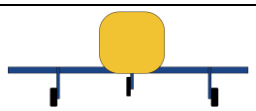
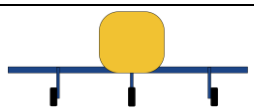
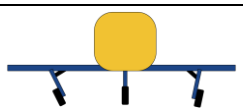
Category	Weight	Single Tractor	Dual Tractor	Single Pusher
Propulsion	**5**	5	4	3
Stability/Control	**5**	4	5	4
Tow Interference	**4**	5	5	2
Structure/Weight	**3**	5	4	3
Historical Record	**3**	5	2	0
Total	**/100**	**95**	**83**	**52**

Tractor, dual tractor, and pusher layouts were considered for the propulsion configuration. The pusher was ruled out early due to the high risk of the tow wire being severed and reduced payload capacity due to the need for excessive ballast forward of the CG. While the dual tractor setup offers potential advantages over a single tractor in top-end thrust, maneuvering performance and low-speed yaw authority, it introduces additional weight, structural complexity, and sources of electrical energy dissipation. Furthermore, a single-motor layout may offer greater clearance for the propeller in a low-wing aircraft, The single tractor configuration was ultimately selected due to the focus of *both* M2 and M3 on cruising efficiency and range over takeoff thrust and top speed, an

unprecedented feature in the recent history of DBF rules.

3.4.7 Landing Gear

Table 3.k: Pugh Chart for Landing Gear Configuration

Category	Weight	 Tail Dragger	 Tricycle	 Retractable Tricycle
Serviceability	**5**	3	5	5
Structure/Weight	**4**	5	4	3
Stability/Control	**4**	3	4	5
Manufacturability	**3**	4	4	4
Aerodynamics	**3**	4	4	5
Total	**/95**	**71**	**81**	**83**

Three conventional landing gear options were evaluated based on payload accessibility, structural weight, ground tracking stability and aerodynamic drag. The tail-dragger configuration was initially considered for its simplicity and weight-saving potential, but was rejected due to poor ground tracking stability, which may trigger ground loops under the unpredictable wind conditions common at the competition site. The incidence angle associated with this configuration also reduces the number of payload access points and the convenience of payload installation/removal, whence the tricycle configuration was preferred. While the nose gear joint constitutes a high-risk failure point during landings amidst gusts/crosswinds, the benefits to ground handling and serviceability were deemed to be decisive. To improve the aerodynamic efficiency of the tricycle layout, especially under high-speed cruising conditions in M2, the team considered installing retractable gear. Following initial weight calculations and early prototype testing, the team decided to install modified commercial off-the-shelf (COTS) units to allow seamless integration into the wing structure, while replacing the shock-absorption struts with strategically bent high-carbon spring steel wire to minimize the empty weight addition to the airframe.

3.4.8 Empennage

Table 3.l: Pugh Chart for Tail Configuration

Category	Weight	Standard	H-tail	T-tail	V-tail
Stability/Control	**5**	4	5	5	2
Structure/Weight	**5**	5	3	3	4
Tow Interference	**4**	4	4	5	5
Manufacturability	**3**	5	3	3	2
Historical Record	**3**	5	2	2	0
Total	**/95**	**86**	**68**	**72**	**52**

Empennage options were evaluated to ensure sufficient longitudinal and directional stability while minimizing weight and mitigating potential interference with the banner tow wire. The team studied four configurations: conventional, H-tail, T-tail, and V-tail. The V-tail and H-tail were ruled out early due to their manufacturing complexity and lack of sufficient historical precedent in similar DBF competitions. These major disadvantages outweighed the minimal gain in pitch stability/control for the H-tail, whose vertical fins act as endplates, and the minimal reduction to wetted area and interference drag for a V-tail, which combines the functions of the vertical

and horizontal stabilizers. The standard tail offers the most straightforward path for manufacturing and assembly and provides reliable stability and control across most flight envelopes. However, a major concern is the potential for the banner tow wire to interfere with the horizontal stabilizer during competitive high-alpha turns, which is not the case for the T-tail. However, superior structural efficiency, manufacturing simplicity, and historical success ultimately yielded the conventional tail as the configuration of choice.

3.5 Initial Sizing Estimate

The results of the score sensitivity analysis, configuration trade studies, and historical data surveys were utilized to project an early sizing estimate for *AARON*, as shown in Table 3.m. The surveys focused on successful aircraft in the 2020 and 2024 DBF fly-offs [2] due to the similar requirements governing internal payloads. Critical quantities surveyed include weight fractions, maximal wing loading, banner length, and fuselage geometric proportions (fineness ratio and width-to-wingspan ratio). Higher wing aspect ratios and wing loadings were favored due to the increased importance of cruising efficiency over takeoff/climb performance as per the mission requirements, and higher banner lengths were favored due to advancements in hobby-grade propulsion technology since 2020. The chief engineer communicated approximate aircraft specifications and payload targets to the technical sub-teams, to enable early conceptualization of the structural layout and payload mechanisms.

Table 3.m: Initial Sizing Estimates

Battery Capacity	99.9 Wh	**Fuselage Width**	8.25 in	**# Passengers**	48	**Banner Length**	25 ft	**Wing Aspect Ratio**	4.25
Wingspan	5 ft	**Fuselage Length**	60 in	**# Cargo**	16	**Gross Weight**	16.2 lb	**Wing Loading**	2.75 lb/ft^2

4. Preliminary Design

To improve upon the initial sizing estimates, the team developed a novel multidisciplinary design optimization (MDO) framework using industry experience, faculty guidance, and scholarly research. In order to produce a robust preliminary design, the framework systematically integrated (i) prior knowledge of classical subsystem-level analytical tools, (ii) expert opinions from experienced prior team pilots and engineers, and (iii) statistical reduction of surveyed historical data.

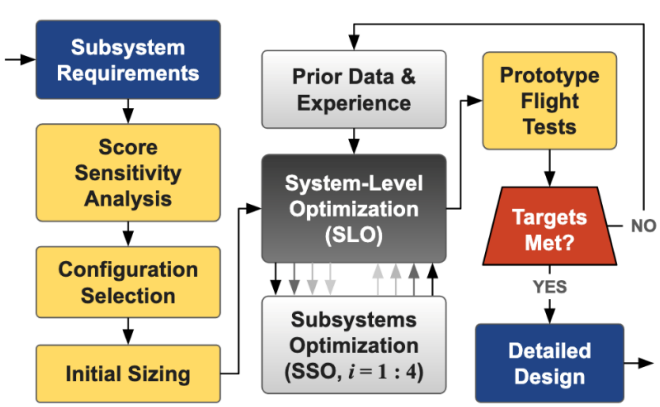


Figure 4.1: Top-Level Block Diagram Overviewing the Team's Preliminary Design Process

Table 4.a: Optimization Problems and Disciplines

Discipline	SLO	SSO			
		$i = 1$ Airfoil Design	$i = 2$ Wing Design	$i = 3$ Aircraft Layout	$i = 4$ Propulsion Selection
Weights	√			√	√
Geometry	√		√	√	
Aerodynamic Prediction	√	√	√	√	
Stability & Control				√	
Propulsion Analysis	√				√
Mission Performance	√				

4.1 Design Methodology

As illustrated using a top-level flowchart in Figure 4.1, the results of the mission requirements decomposition, scoring sensitivity analysis, configuration selection, and initial sizing were fed into the team's MDO framework, whose output was validated through early prototype flight tests. As overviewed using a compact Design Structure Matrix (DSM) [4] in Figure 4.2, the MDO framework exploited a custom-designed, multi-fidelity, distributed optimization architecture, named the Recursive-Refinement Design Optimization (*RRDO*) method. *RRDO* enabled efficient maximization of the ultimate objective $f_0(\boldsymbol{x}_0, \boldsymbol{x}_f)$, the total mission score, through iterative solution of one system-level optimization (SLO) problem and four subsystem-level optimization (SSO) problems. The SSO's supplemented the SLO by improving the level of precision and fidelity in discipline-specific analyses, as outlined in Table 4.a.

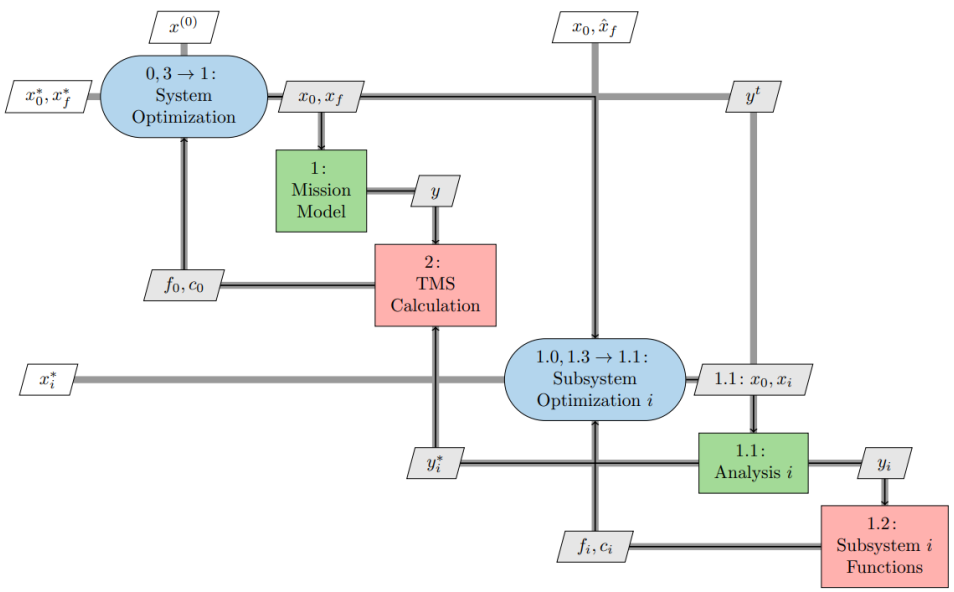


Figure 4.2: Compact DSM Depicting *RRDO* Distributed Optimization Architecture

At the system level, primary design variables $\boldsymbol{x}_0$ were passed through a low-fidelity mission model that applied sequential discipline analyses for weight estimation, aerodynamic prediction, and mission performance evaluation, as detailed below in §4.2.1. Design variables that explicitly correlate with f_0 (as seen in Eq. 3.5–3.9) informed top-level sizing and gross weight estimates. Auxiliary design variables including maximal wing loading W_o/S and wingspan b, as well as maneuver-specific performance targets $\boldsymbol{x}_f$, were integrated using physics-based semi-empirical models to compute important state variables $\boldsymbol{y}_i$, such as aerodynamic efficiency L/D and required electrical power P (§4.2.2). The design space was strategically reduced by imposing constraints derived from trade studies as well as historical statistics that were extrapolated using principles of Bayesian inference (§4.2.3, §4.2.4). The system-level optimization produced practical sizing and performance estimates $[\boldsymbol{x}_0^*, \boldsymbol{x}_f^*]$, along with uncertainty intervals that account for epistemic variabilities introduced by the low-fidelity predictive models utilized (§4.2.5). Following the initial system-level optimization, four subsystem-level "suboptimizations" were run in sequence using higher-fidelity modeling tools, to progressively refine the detailed aerodynamic outer mold line and specific propulsion package (§4.3, §4.4). Each suboptimization was performed in two stages: (i) maximization of a discipline-specific, subsidiary objective $f_i(\boldsymbol{x}_i)$ with respect to local design variables $\boldsymbol{x}_i$ subject to appropriate

constraints $\boldsymbol{c}_i(\boldsymbol{x}_i) \leq 0$, followed by (ii) system-level re-optimization of the performance targets $\boldsymbol{x}_f$ by integrating higher-fidelity computation of relevant linking (state) variables $\boldsymbol{y}_i \subset \boldsymbol{y}$ into the mission model. This decomposed, recursive, multi-fidelity approach enabled efficient exploration of the multi-dimensional design space while ensuring the timely delivery of an optimized preliminary aircraft design that meets all mission requirements.

4.2 System-level Sizing Optimization

4.2.1 Mission Model

A multi-mission predictive model was assembled using semi-empirical equations that allow efficient computation of the mission scores $[S_{GM}, S_{M1/2/3}]$ as functions of the primary design variables. The servicing time t_{UCSD} in the GM was expressed explicitly in terms of payload quantity $[N_C, N_P, L_{\mathrm{Bann}}]$ by fitting a linear function to data collected from timed tests using preliminary payload-system prototypes, as exemplified in Figure 4.3c.

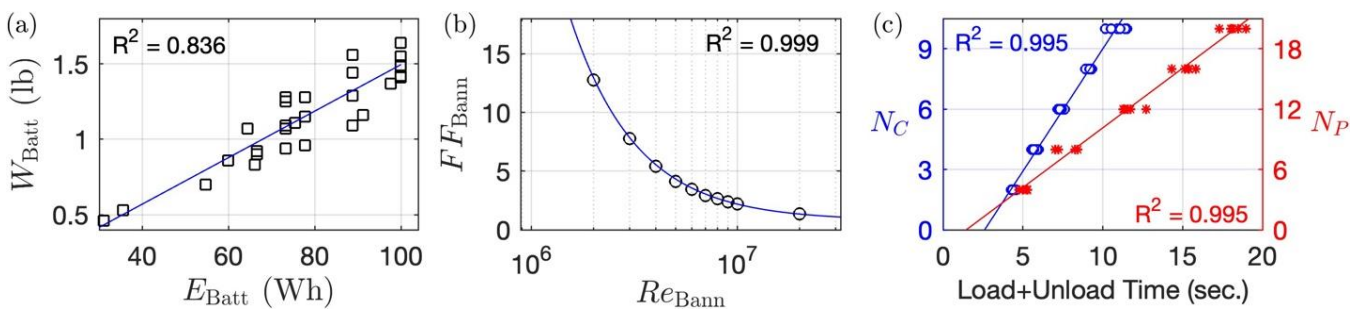


Figure 4.3: Empirical Models: (a) Battery Energy Density, (b) Membrane Form Factor, and (c) Payload Servicing Time

The expected lap count N_L in each competitive flight mission (M2/M3) was computed as an implicit function of payload quantity, wing size, available electrical potential energy, and desired flight performance metrics. The function incorporated (i) equilibrium equations that represent the precise dynamics of each energy- and time-intensive maneuver (takeoff, climb-out, cruise, level turns), (ii) empirical models that exploit practical data to inform the prediction of auxiliary quantities (battery energy density, propulsive/structural efficiencies, time-averaged drag of a fluttering membrane [2], weight of a ballast-equipped banner, etc.) without incurring undue computational costs (see, for example, Figures 4.3a and 4.3b), and (iii) factors of safety (FoS) that limit usable electrical energy, usable time, and aerodynamic efficiency, to ensure robustness of the optimal design in the face of severe uncertainties posed by the unstable winds/gusts anticipated at the competition site [6] (see Figure 4.4). The empirical models and FoS entered through a set of fixed parameters $\boldsymbol{p}_0$ in the system-level problem. Of particular importance was capturing the variation of aircraft weight and drag with the amount of internal payload $[N_P, N_C]$. For this purpose, the team surveyed empty weight fractions and geometrical proportions of successful aircraft in prior DBF competitions with similar high-volume payload requirements. A mathematical model was derived to predict fuselage dimensions in terms of net payload storage volume, whence the excess parasitic drag was computed using the form factor method [5].

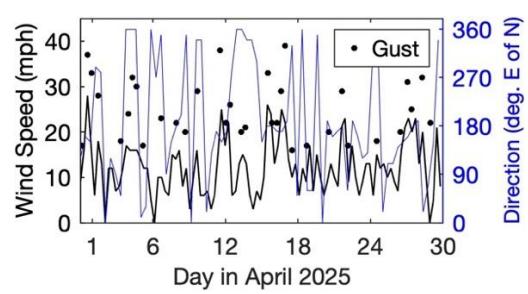


Figure 4.4: Sampled Fly-off Site Wind Data

4.2.2 System-level Optimization Problem

The system-level problem involves maximizing a modified form of TMS that assumes completion of all missions,

$$f_0(\boldsymbol{x}_0, \boldsymbol{x}_f\,; \boldsymbol{p}_0) = 4 + \frac{1/t_{\text{UCSD}}}{1/t_{\min}} + \frac{[\text{Net Income}]_{\text{UCSD}}}{[\text{Net Income}]_{\max}} + K_W \frac{[N_L L_{\text{Bann}}/\text{RAC}\,]_{\text{UCSD}}}{[N_L L_{\text{Bann}}/\text{RAC}]_{\max}}, \tag{4.1}$$

with respect to design variables representing payloads, battery capacity, wing size, and aerodynamic efficiency, $\boldsymbol{x}_0 = [N_C, N_P, E_{\text{Batt}}, L_{\text{Bann}}, W_o/S, b]$, and maneuver-specific performance variables $\boldsymbol{x}_f$ that are shared with the suboptimization problems. The Pareto coefficient K_W is introduced as a random variable to model the disproportionate risk of performance overestimation in M3 due to both epistemic and aleatory uncertainties, as discussed in §4.2.4. The design space is subject to the constraints $\boldsymbol{c}_0(\boldsymbol{x}_0) \leq 0$, including bound constraints that enforce compliance with payload rules (see Table 3.a) and reduce computational cost, and nonlinear constraints that enforce the federal gross weight limit and ensure that the wing lift coefficient remains safely below its maximum value during all maneuvers. (Variables with arbitrarily fixed bounds were carefully monitored to ensure validity of the solution.)

The nonlinear maximization problem defined above was implemented in the MATLAB software using a combination of the interior-point and sequential-quadratic-programming algorithms, both backed by a scatter-search routine for effective diversification of seed states beyond the crude initial sizing estimate. Within the gradient-based environment, the cargo/passenger quantities $[N_C, N_P]$ and lap count N_L were treated as pseudo-continuous variables in the first pass of optimization and sequentially rounded for physical validity. Results were validated within MATLAB using a genetic algorithm tailored for integral design variables and a non-smooth solution topography.

4.2.3 Solution Strategy

The fundamental issue driving system-level design was determining the best strategy for allocation of priority between the various missions, to maximize the likelihood of scoring among the upper echelon of teams at the fly-off. The team's structured decision-making approach involved five key factors:

I. Quantifying the tradeoffs between different missions that involve conflicting subsystem requirements. This analysis was systematized by using the MDO solver to produce a sensitivity matrix, as shown in Table 4.b. Each cell in the matrix identifies the salient physical considerations underlying the low or high sensitivity.

II. Prediction of the top score in each mission, using the MDO solver and historical data surveys.

III. Projection of the team's ability to compete against the top-scoring teams in each mission. The analysis employed a matrix that enumerates mission-critical skills and assets, as shown in Table 4.c.

IV. Quantification of aleatory uncertainties yielding a high risk of score reduction in each mission. In particular, the analysis focused on assessing disproportionate risks in M3 due to strong winds, as shown in Table 4.d.

V. Statistical analysis of historical trends in priority allocation and performance at prior DBF competitions.

Table 4.b: Mission Sensitivities Correlation Matrix

Variable	GM	M2	M3
$N_P + N_C$	t_{UCSD}	Net Income	$C_{D,i}$
N_P/N_C	t_{UCSD}	$C_{D,i}$	$C_{D,i}$
L_{Bann}	t_{UCSD}	$L/D, \eta$	S_{M3}
b		L/D	RAC
W_o/S		P, N_L	P, N_L
E_{Batt}		EF	N_L
Sensitivity Color Code: >> 0 | > 0 | < 0 | << 0			

Table 4.c: Important Skills/Assets Matrix

Capability/Resources	GM	M2	M3
Intelligent mechanism design	√√	√	√
Servicing speed & consistency	√√		
Efficient structural design		√√	
Adequate budget		√	√√
Test/aerobatic piloting skill		√√	√
Reliable & light manufacturing		√√	√
Aero-propulsive optimization		√	√√
Aerodynamics testing facility			√√

The team's solution approach coupled two distinct tools: (i) trade studies that employed the system-level MDO solver, and (ii) strategic inference from historical data and experience using principles of Bayesian game theory [7]. The design process for the 2025–2026 DBF competition was modeled as a simultaneous, non-cooperative, interim-stage Bayesian game. The game is subject to highly incomplete information regarding player strategies and feasible payoffs, but most likely involves common knowledge arising from multidisciplinary trade studies (among skilled players). The 2026 fly-off itself was identified as a sequential but strongly asymmetric game, due to moves by nature – most prominently, variations in wind/gust levels through the slow flight-order rotation. The lack of information warranted a Bayesian approach to model player type profiles and action sets, drawing on prior probability distributions constructed from data in multiple prior competitions: those with similar mission requirements and scoring functions. The effects of wind were modeled using the random variable K_W, whose moments were determined using probabilistic analyses of prior meteorological data.

Table 4.d: M3 vs. M2 Disproportionate Environmental Risk Analysis

Source	Greater Risk(s) in Mission 3
Lower W/S	• Increased gust sensitivity through the mass parameter
Lower q_∞	• Reduced passive stability and control response
High L_{Bann}	• Increased control interference amidst gusts • Increased trim drag during turns due to towing moment • Increased epistemic uncertainty in net drag prediction
Lower V_{CR}	• Increased gust sensitivity due to gust-induced AoA and gust-induced airspeed inflation • Greater relative increase in drag, $2V_W/V_{CR} + (V_W/V_{CR})^2$ for a given headwind speed V_W

The five-pronged approach outlined above revealed a clear bifurcation in possible strategies, based on the central sizing variables: passenger/cargo count. Prioritizing GM would reduce the effort required for design and execution but place the team at the risk of scoring poorly due to the entry of teams with superior mechanical design experience and payload servicing training. On the other hand, prioritizing M2 severely increases the difficulty level of detailed design and reliable manufacturing, while placing the team at a much lower risk of score reduction due to wind/gust interference, as explained in Table 4.d. As detailed below, a robust process that integrates Bayesian inference, multidisciplinary optimization, and expert opinions was utilized to clarify and solve this central problem.

4.2.4 Solution Process

The solver was first deployed to produce intelligent predictions for the maximum mission scores expected at the 2026 DBF fly-off. For GM, it was deduced using the historical data that at least one team would opt for the minimum legal payload amount to maximize payoff while incurring the least possible difficulty in design and manufacturing. Results of the M2/M3 analyses are visualized in Figure 4.5, where the maximum possible scores are plotted as a function of mission-specific payload variables. In both missions, a tradeoff was observed between the chosen payload amount and the maximum number of laps N_L that may be completed given the limited time and potential energy. In M3, this tradeoff led to a clear optimum for the banner length, beyond which value the decrease in range/endurance due to excessive net drag penalized the mission score. A similar trend was observed for the cargo count N_C in M2, as anticipated based on drag scaling arguments. On the other hand, the M2 score was observed to grow monotonically with the number of passengers N_P. To close the problem, the team thus collected and analyzed historical statistics of fuselage sizes and payload weight fractions. Note that, to yield conservative predictions for GM/M2/M3 that account accurately for the high skill level of top teams, the values of specific empirical parameters influencing structural, aerodynamic, propulsive, and servicing efficiencies were increased from their base values using data obtained from analogous surveys.

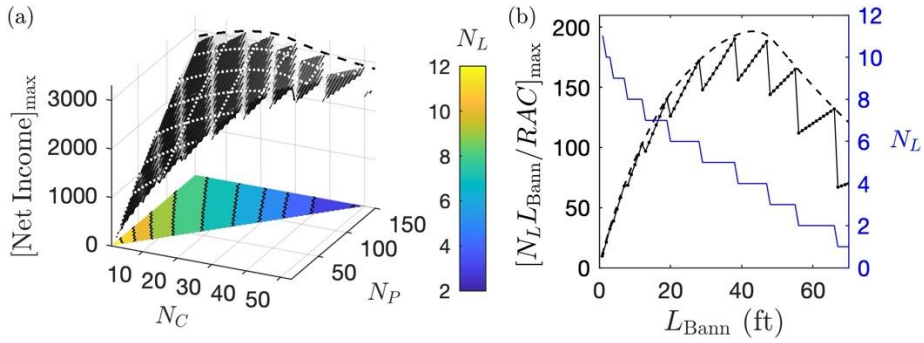


Figure 4.5: Maximum Score MDO Trade Studies for (a) M2 and (b) M3

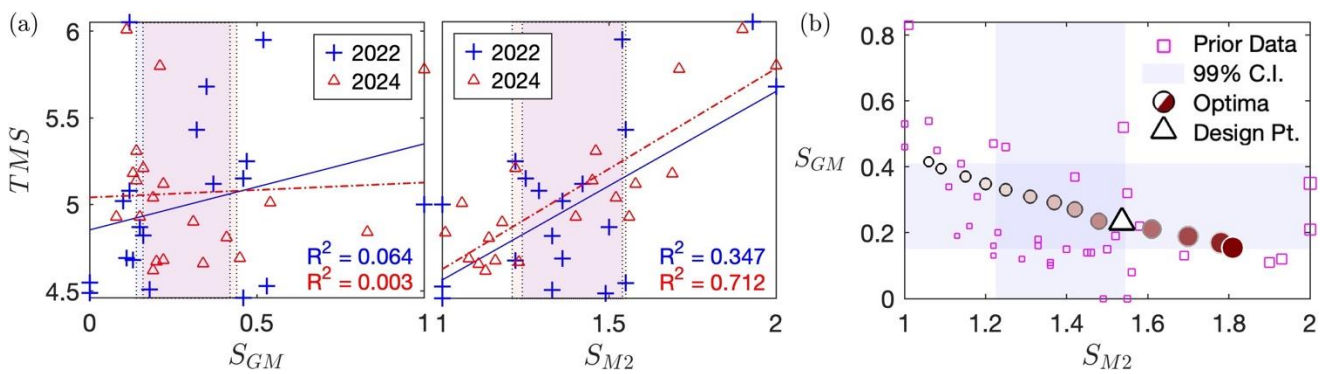


Figure 4.6: (a) Historical Score Correlation Statistics and (b) Data-contextualized Pareto Front for Payload Selection (where Marker Size Represents TMS and Shade Represents $N_P + N_C$)

Selection of internal payload, central to sizing, was conducted by analyzing historical correlations between TMS and GM/M2 scores among the highest-performing twenty teams in the similar 2022 and 2024 DBF competitions [2]. As shown in Figure 4.6a, the statistics of correlation display remarkable similarity between the two datasets despite minor differences in the scoring formulae and payload types, suggesting that they serve as a strong prior for Bayesian inference of the optimal passenger/cargo count. As shown in the data-overlaid Pareto front in Figure 4.6b, 95% confidence intervals computed from the data were used to generate inequality constraints for S_{GM} and S_{M2}. While the constrained design point fell short of the single non-dominated solution, indicated by the largest and darkest circle, it was selected as the optimal compromise between GM servicing efficiency and M2 'Net

Income' – that which maximizes the likelihood of scoring among top teams in the 2026 fly-off.

Having frozen N_C and N_P, the MDO algorithm was utilized to investigate the recommendations produced by the initial score sensitivity analysis for maximizing the wingspan (b) and battery capacity (E_{batt}). Trade studies revealed that reducing b to benefit S_{M3} through the RAC and reducing E_{batt} to` benefit S_{M2} through the EF yields a maximum increase in TMS of under 1%. The team only considered values of E_{batt} corresponding to COTS lithium-ion polymer batteries (due to their high energy density and discharge capability) sold by a wide selection of reputable manufacturers. As shown in Figure 4.7 by comparison to historical statistics, this minute penalty is unlikely to substantially lower the rank of the team – assuming placement within the top ≈25 and neglecting differences in the total report score. Furthermore, prior team experience revealed that the consequences of compromising endurance (through aerodynamic efficiency in M2 and electrical potential energy in M3) may yield catastrophic results in the face of severe winds/gusts that may inflate cruise power consumption and/or force aborted landings. Thus, equality constraints were applied, setting $b = 5$ ft and $E_{\text{batt}} = 99.9$ Wh.

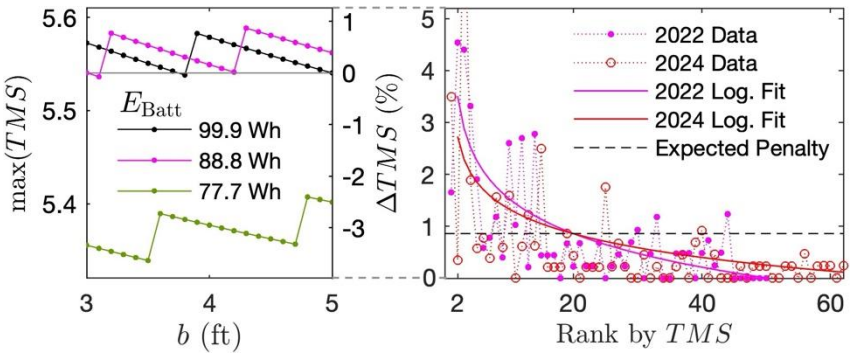


Figure 4.7: Data-backed Analysis of TMS Penalty Due to EF and RAC

The above results were found to be effectively insensitive to the value of the gross wing loading W_o/S. The gross weight W_o having been deduced from $[N_C, N_P, E_{\text{batt}}]$, and the span b having been frozen, W_o/S mainly correlated with the aerodynamic efficiency L/D of the aircraft through the wing aspect ratio AR. The team noted that both M2 and M3 reward the discrete number of laps completed, N_L, as opposed to the time taken to complete a given number of laps, for the first time in six years of DBF competitions with otherwise similar ruling frameworks. In the absence of a continuous performance metric influencing the objective, the MDO algorithm proposed wide ranges of wing loadings that achieved identical scores through optimization of the flight maneuvers $\boldsymbol{x}_f$. Internally, the gradient-based optimizer balanced speed and power consumption in such a way as to equalize the non-integer number of laps $\mathcal{N}_L$ that are feasible within the legal limits on time and energy. Note here that $N_L = \lfloor \mathcal{N}_L \rfloor$. To resolve this non-uniqueness, the team identified the continuous lap count $\mathcal{N}_L$ and maximum power consumption P as the salient selection criteria for W_o/S. The former measures excess range/endurance, which increases the likelihood of completing the desired N_L amidst reducible and irreducible uncertainties, i.e. model inaccuracies and wind/gust interference. The latter correlates with the cost, weight, and power rating of the required motor, which may increase the induced drag penalty and compromise propulsive efficiency under cruising conditions.

The MDO algorithm was used to produce a plot of the required power-to-weight ratio P/W_o as a function of W_o/S, as shown in the lower panel of Figure 4.8. The blue curves corresponding to the four critical steady maneuvers in M2, and the red curve represents the maximum power envelope for M3 with a sample banner length of 20 ft. Each data point in this constraint diagram represents an aircraft whose flight performance has been optimized using gradient-based methods to maximize $\mathcal{N}_L$. As shown in the upper panel of the figure, $\mathcal{N}_L$ reaches global maxima for highly disparate wing loadings across the two missions.

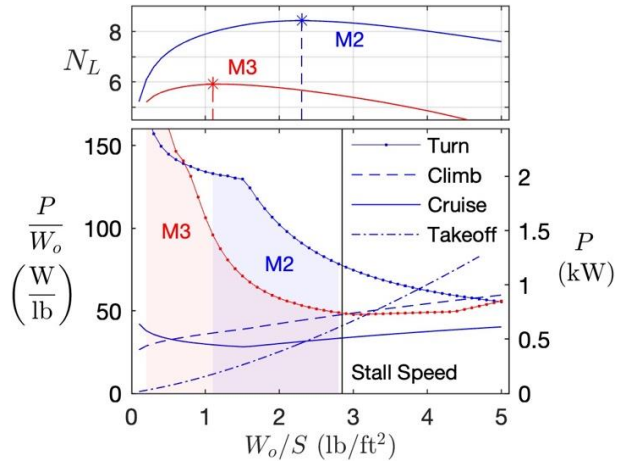


Figure 4.8: Optimal Propulsion Constraint Diagram

Higher wing loadings were preferred due both to conventional recommendations regarding P/W_o selection for model sport aircraft [8] and to the disproportionate vulnerability in M3 to moves by nature (see Table 4.d). In pursuit of a structured compromise, the team adapted fundamental principles of stochastic MDO [9]. The random variable K_W , a Pareto coefficient scaling M3 performance in the objective (Eq. 10), was assigned specific moments that represent the detrimental effects of headwind-induced banner drag inflation and gust-induced control interference. The moments were formulated as functions of L_{Bann} through comparison of the optimal maneuvering speeds $\boldsymbol{x}_f$ and the anticipated wind/gust speeds U. The function drew on strong priors in the form of well-suited Weibull probability density functions (PDFs) fitted to meteorological data from the competition site, as detailed in Figures 4.9a and 4.9b. A Monte Carlo analysis was conducted, using as the objective a modified form of TMS where N_L was replaced with $\mathcal{N}_L$. Such a formulation rewards excess range and endurance in M2 and M3, while measuring their relative importance using K_W and the original maximum score predictions. The resulting posterior PDFs for the optimal wing loading $(W_o/S)^*$ are shown in Figure 4.9c for a wide range of relevant banner lengths. The increasing level of asymmetric leptokurtosis for higher values of L_{Bann} demonstrated that higher wing loadings are preferable not only in M2 for aero-propulsive efficiency but also in M3 for robustness to various wind conditions and banner lengths. The latter was deemed of great importance since the rules allow the team to select between two banners of different lengths. Using the results of the study and consultation with experienced pilots, the team selected a robust, practically favorable wing loading and used the optimizer to project the maximum desired banner length. Results of the system-level optimization are enumerated below, which was imposed through a final equality constraint.

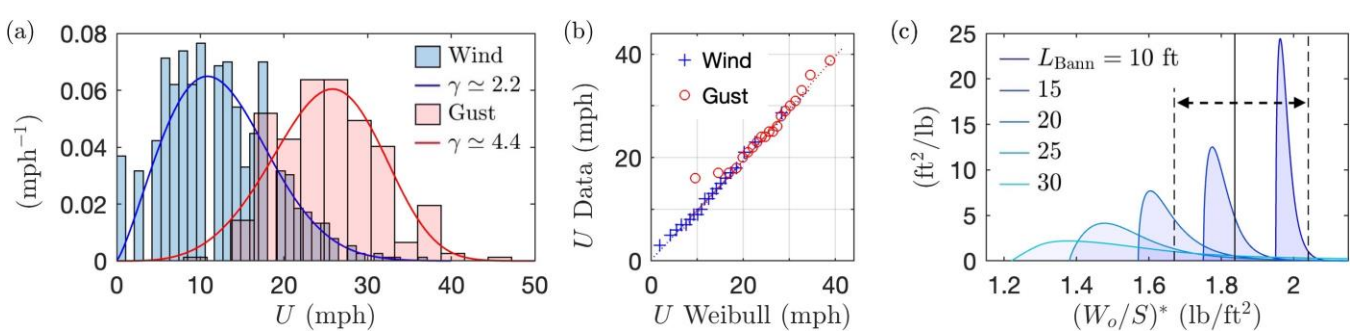


Figure 4.9: (a) Prior Probability Distributions of Wind/Gust Speeds at Fly-off Site, (b) Quantile–Quantile Assessment of Weibull Regression, and (c) Expected Probability Distributions of Optimal Wing Loading for Various Banner Lengths

4.2.5 System-Level Sizing and Performance Targets

Upon incorporation of the deduced scoring constraints, sizing constraints, and robust wing/banner sizing, the total mission optimizer was deployed to compute the system-level optimum $[\boldsymbol{x}_0^*, \boldsymbol{x}_f^*]$. To improve the robustness of the subsystem-level design decisions that were to follow, a sensitivity analysis was conducted to identify the most influential of the empirical parameters $\boldsymbol{p}_0$ employed in the mission model. Upon analysis of the numerically estimated Jacobian matrix $\partial \boldsymbol{x}_0^* / \partial \boldsymbol{p}_0$, the values of the most consequential parameters were varied by ±10% to produce best- and worst-case predictions for each system-level design variable. Selected results are displayed in Table 4.e. The deterministic solution and its epistemic uncertainty margins were communicated to the technical sub-teams for the development of a versatile outer mold line (OML) and propulsion system.

Table 4.e: System-level Design Solution (black text) with Uncertainty Intervals (blue: best case, red: worst)

Payloads		Sizing		Weights (lb)		Flight Performance	
N_C	12	E_{batt}	99.9 Wh	Empty	7.65 (6.26 – 9.36)	N_L (M2)	8 (7 – 9)
N_P	38	b	5 ft	Battery	1.49	N_L (M3)	5 (4 – 6)
L_{Bann}	28 ft (22.5 – 28.5)	AR	3.0	Banner	1.05 (0.69 – 1.08)	V_{CR} (M2)	83 ft/s (81 – 89)
		W_o/S	1.837 lb/ft^2 (1.67 – 2.04)	Gross (W_o)	15.31 (13.92 – 17.01)	V_{CR} (M3)	47 ft/s (49 – 51)

4.3 Aerodynamic Design

Three subsystem-level optimizations were performed as part of the aerodynamic design process: airfoil design for multi-point efficiency, wing planform refinement for high-alpha performance, and tail configuration for adequate stability. Development of the OML began with the design of a robust lifting surface that provides a competitively high lift-to-drag ratio L/D and maximum lift coefficient $C_{L,\max}$ across the disparate airspeeds spanned by critical maneuvers in the competitive flight missions (M2/M3). The team defined an expanded range of Reynolds numbers $Re \in \mathcal{R}$ for analysis, which incorporated pre-established uncertainty ranges for atmospheric properties and maneuver-specific airspeeds, as well as a range of taper ratios permitted under structural considerations.

4.3.1 Airfoil Design (SSO $i = 1$)

The airfoil optimization problem involved maximizing a linear combination of state variables given by

$$f_1(\boldsymbol{x}_1) = \sum_{j=2,3} \left[k_1 C_{L,\max,\text{TU}} + k_2 L/D_{\text{CR}} - k_3 C_{M,\text{CR}} - k_4 \alpha_{\text{CR}}\right]_{Mj}, \tag{4.2}$$

with respect to a local variable $\boldsymbol{x}_1$ that represents the finite set of spatial coordinates defining an airfoil, subject to qualitative constraints pertaining to manufacturability and geometric constraints that strategically reduce the design space as explained below. The subsidiary objective f_1 evaluates airfoils based on four key performance criteria in each of the two competitive flight missions (M2/M3): maximum lift coefficient at the turning airspeed ($C_{L,\max,\text{TU}}$), and lift-to-drag ratio, quarter-chord pitching moment coefficient, and angle of attack (AoA) during cruise ($L/D_{\text{CR}}, C_{M,\text{CR}}, \alpha_{\text{CR}}$). The penalty coefficients k_3 and k_4 were included to model the hidden tradeoff between the trim-drag/tail-weight penalty, required to combat the strong downward moment typically generated by rear-cambered high-lift airfoils, and the excessive induced drag generated by a high-volume fuselage at a nonzero AoA, which accompanies weakly cambered airfoils that are mounted with zero incidence. (Based on conversations with the Structures and Fabrication sub-teams, wing incidence was avoided for the sake of simplicity in the design and

manufacturing of a self-locating fuselage–wing–undercarriage assembly.)

The strongly 3-dimensional aerodynamics of low-aspect ratio wings warrants a holistic design methodology that closely couples spanwise and chordwise flow analysis. Thus, the team's airfoil optimization process integrated viscous vortex-panel flow simulations of rectangular wings with the design aspect ratio $AR = 3$. The design space was first reduced by imposing geometrical constraints derived from historical data surveys and preliminary simulations of high-efficiency airfoils (obtained from the digital 'Airfoil Tools' database [10]) in the XFLR5 software for relevant values of $Re \in \mathcal{R}$. In particular, the maximum thickness was constrained between 9.5% and 12% to minimize profile drag while ensuring sufficient internal volume for load-bearing structure, and the maximum camber was constrained between 2% and 4% to enable sufficiently high values of $C_{\ell,\max}$ (i.e. safe stall speeds) while avoiding the degradation of L/D within the specific window of design lift coefficients $C_{L,\,\mathrm{CR}}$ in M2 and M3. Filtration of the 1,648 airfoils in the database thus yielded 140 candidates, from which 19 were selected for further analysis based on manufacturability criteria stipulated by experienced team members and advisors.

Table 4.f: Abridged Airfoil Selection Matrix Based on 3D Flow Simulation

Metric	Mission 2				Mission 3				Net Score
	$C_{L,\max,\mathrm{TU}}$	L/D^*	C_M^*	$\alpha^*(°)$	$C_{L,\max,\mathrm{TU}}$	L/D^*	C_M^*	$\alpha^*(°)$	
Scale	*1.362*	*17.70*	*–0.09*	*–1.08*	*1.29*	*15.64*	*–0.08*	*3.81*	***/ 2***
Weight	**20%**	**80%**	**–2%**	**–10%**	**20%**	**80%**	**–2%**	**–10%**	
SD7032	1.36	16.96	–0.08	–0.84	1.28	15.83	–0.07	4.04	1.757
FX60-100SM	1.34	17.93	–0.10	–1.30	1.28	15.49	–0.09	3.60	1.739
TR26	1.37	17.50	–0.09	–1.03	1.29	15.71	–0.08	3.86	1.759

The short-listed airfoils were evaluated based on all criteria listed in Eq. 4.1, whence the Selig/Donovan *SD7032* and the smoothed Wortmann *FX60-100SM* emerged as optimal candidates for M2 and M3, respectively. The Wortmann airfoil generally demonstrates superior L/D_{CR} at high airspeeds, while the Selig/Donovan offers critical advantages in low-speed handling qualities – specifically, a gentler stall behavior due to its leading-edge geometry. The *SD7032* was also deemed to be more favorable from the standpoint of structurally efficient manufacturing due to its larger trailing-edge wedge angle. To achieve a profitable compromise, the team conducted a blending optimization study where various linearly interpolated airfoils $\boldsymbol{x}_1$ were evaluated using the discipline-specific objective J_1, as exemplified in the weighted selection matrix in Table 4.f. The team converged on a 40%-*FX* / 60%-*SD* blend as a suitable compromise between manufacturability and multi-mission suitability. The performance of the synthesized airfoil, designated the "Triton" *TR26* in honor of the university mascot, is compared with that of its parent airfoils in Figure 4.10. Following finalization of the airfoil, improved predictions for the state variables $\boldsymbol{y}_1 \subset \boldsymbol{y}$ representing lift and drag polars were incorporated into the system-level optimizer, and refined values for the maneuver-specific performance targets $\boldsymbol{x}_f$ were passed to the wing geometry suboptimization problem.

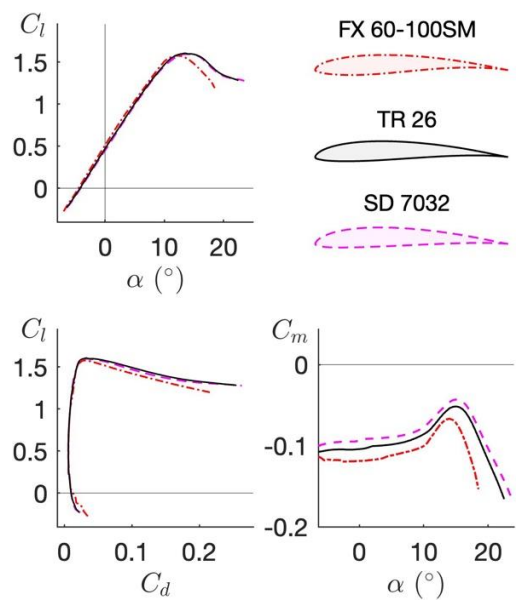


Figure 4.10: Airfoil Polar Plots ($Re = 7.5e5$)

4.3.2 Wing Geometry Optimization (SSO $i = 2$)

Following finalization of the airfoil, the team conducted a geometrical optimization study aimed at further improving the cruising efficiency and stall behavior of the low-aspect-ratio wing. A sensitivity analysis conducted using XFLR5 demonstrated that modifying the quarter-chord sweep angle, taper ratio λ, and geometric twist (washout) altered L/D_{CR} negligibly (by the order of 1%). As argued in scholarly literature, since $AR \lesssim 4$, the lift distribution is approximately elliptical – and, hence, nearly optimal for a limited wingspan – regardless of planform geometry [11,12]. On the other hand, $C_{L,\max}$ was found to increase aggressively (order of 10%) in response to decreasing taper ratios, till a critical threshold beyond which it decayed due to the tip Reynolds number becoming restrictively small and triggering early laminar separation. The atypical increase occurring for small/moderate values of $1 - \lambda$ was attributed to an increase in the root Reynolds number that delayed an inboard stall, the latter possibly occurring first due to outboard suppression of boundary-layer separation by the swirling flow dominating the near-tip aerodynamics of the low-AR wing. To further improve stall behavior while preserving simplicity in design/manufacturing, the leading edge was constrained to be linear, thereby providing effective forward sweep that induces favorable spanwise airflow. Such a configuration is commonly used to also reduce roll stability and thus improve maneuverability for full-scale, mid-wing ultra-aerobatic aircraft.

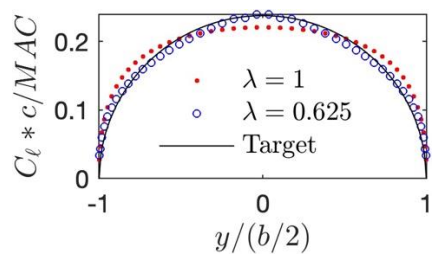


Figure 4.11: Spanwise Lift Distribution

Based on the sensitivity study, the team defined a wing geometry with a rectangular center section and tapered outboard panels: a configuration that has achieved great success in DBF competitions with similar mission requirements [2]. A two-parameter subsystem optimization problem was defined, to maximize

$$f_2(\boldsymbol{x}_2) = \sum_{j=2,3} \left[C_{L,\max,\mathrm{TU}}\right]_{Mj} \text{ with respect to } \boldsymbol{x}_2 = [\lambda, b_i/b] \,, \tag{4.3}$$

where b_i/b denotes the inboard span fraction. Viscous flow simulations were conducted at the updated turning airspeeds in M2/M3, using an in-house Python code that interfaced with the XFoil and AVL (Athena Vortex Lattice) software. As exemplified using a comprehensive contour plot in Figure 4.12, b_i/b was subject to an inequality constraint to ensure sufficient aileron authority without inducing excessive maneuvering drag, and λ was constrained through a minimum value for the tip chord c_t, to ensure sufficient internal volume for load-bearing structure. These constraints drove the team to converge on a design point of $\{\lambda = 0.625,\ b_i/b = 0.1\}$, which yielded improvements in $C_{L,\max,\mathrm{TU}}$ of 12% in M2 and 11% in M3. The spanwise lift distribution of the optimized wing at the M2 cruise condition is compared in Figure 4.11 with that of the rectangular wing. Following finalization of the wing design, improved predictions for state variables $\boldsymbol{y}_2 \subset \boldsymbol{y}$ representing lift and induced drag were incorporated once again into the system-level optimizer, to produce refined values for the maneuver-specific performance targets $\boldsymbol{x}_f$.

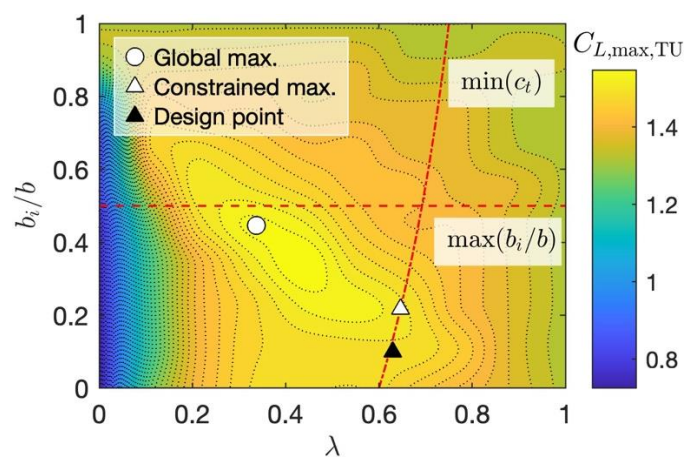


Figure 4.12: Wing Planform Design Space (M2)

4.3.3 Aircraft Layout Optimization (SSO $i = 3$)

Based on the results of the initial sizing (Table 4.e) and the unique requirements for internal payload storage (Tables 3.a), the team recognized that the design of the fuselage, tail surfaces, and wing mount were inherently connected in a coupled optimization problem for aerodynamic efficiency and stability in M2. (M3 was considered of lower importance in this context due to the disproportionate drag contribution from the towed banner and the far smaller payload weight fraction.) Sizing a streamlined fuselage around the slender passenger compartment and compact cargo hold (see Table 3.f), fixing the longitudinal wing location to provide a sufficient static margin in all three missions, and simultaneously sizing the tail surfaces to achieve reasonable tail volumes yielded a severely short-coupled aircraft prone to pitch control issues such as pilot-induced oscillations (PIO) in the absence of active stabilization systems. Preliminary calculations showed that shifting the wing forward would require the addition of ballast to maintain static pitch stability, thereby reducing the M2 score through payload capacity or range, while extending the fuselage to achieve a reasonable tail arm length may increase weight and/or cruising drag in both M2/M3 as well as dampen pitch authority due to a restrictively high fuselage inertia.

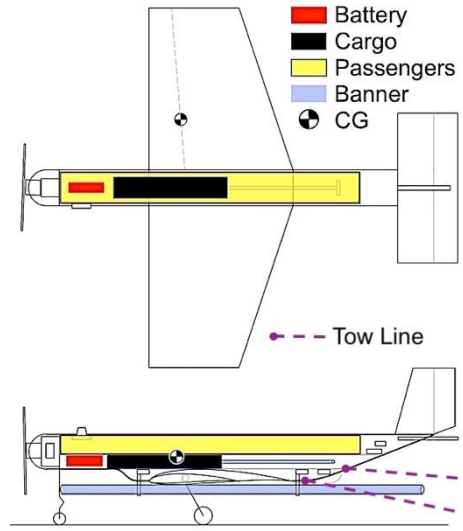


Figure 4.13: Preliminary Layout

Table 4.g: Low-Fidelity Layout Constraints

Lift Force	W_o	Static Margin	20 %
Parameter		Min. : Max.	Design
TVC_H		0.35 : 0.75	0.40
TVC_V		0.07 : 0.20	0.15
Tail Arm / MAC		1.75 : 2.25	1.91
Fineness Ratio		8.5 : 11.5	9.84

The team thus formulated a lift-constrained, static-margin-constrained, drag-minimization problem that employed gradient-based methods to simultaneously optimize the fuselage length L_F, wing position x_W, and horizontal/vertical tail planform areas $[S_H, S_V]$, to maximize cruising efficiency while providing favorable handling qualities. The objective, $f_3(\boldsymbol{x}_3 = [L_F, x_W, S_H, S_V]) = (C_{D,\mathrm{CR}})_{M2}$, was defined as the sum of parasitic drag across the fuselage, wing and empennage, and the additional induced drag due to pitch trim—all under the M2 cruise condition. In addition to the equality constraints on net lift and static margin (SM), the design variables were subject to two sets of inequality constraints $\boldsymbol{c}_3(\boldsymbol{x}_3) \leq 0$ representing practical stability criteria: $\boldsymbol{c}_3^L \subset \boldsymbol{c}_3$, geometric restrictions derived from a parametric survey of successful DBF aircraft, as exemplified in Table 4.g, and $\boldsymbol{c}_3^H \subset \boldsymbol{c}_3$, general-aviation standards for static stability derivatives and military standards for dynamic-mode eigenvalues. The survey prioritized aircraft with similar internal volume to account for the effects of a large fuselage on pitch inertia, form drag and, crucially, yaw stability due to side area. The constraints $\boldsymbol{c}_3^H$ were applied outside of the loop since they require a disparately high level of computational fidelity, as detailed in §4.3.4. The objective f_3 and lower-fidelity constraints $\boldsymbol{c}_3^L$ were computed efficiently using semi-empirical formulae derived using physical principles, DBF data surveys, and numerical experiments involving boundary-element aerodynamic simulations, as outlined in Table 4.h. Sample results are visualized in Figure 4.14, where the minimum achievable nominal cruising drag is plotted as a function of L_F and x_W (measured in inches aft of the foremost fuselage bulkhead). Note here that the NACA 0008 airfoil was chosen for the tail surfaces as a compromise between cruising efficiency, deep-stall/rudder-lock resistance, and structural resistance to gust-induced aerodynamic loads.

Table 4.h: Layout MDO Efficient Computational Framework

Values	Variables	Computation Method
SM	Aerodynamic Centers of Wing & Stabilizers	Data from Numerical Experiments in AVL, & Equilibrium Equations
TVC_H	Aircraft Neutral Point	
TVC_V / **Tail Arm**	Aircraft Center of Gravity Location	Itemized Weight Breakdown using Data Surveys of Prior DBF Reports & RC Products
Net Drag	Induced & Parasitic Drag, and Lift: Wing & Horizontal Tail	Data from Numerical Experiments in XFLR5, & Equilibrium Equations
Net Lift	Parasitic Drag: Fuselage & Vertical Tail	Empirical Form Factor Method: Geometry & Equivalent Skin Friction

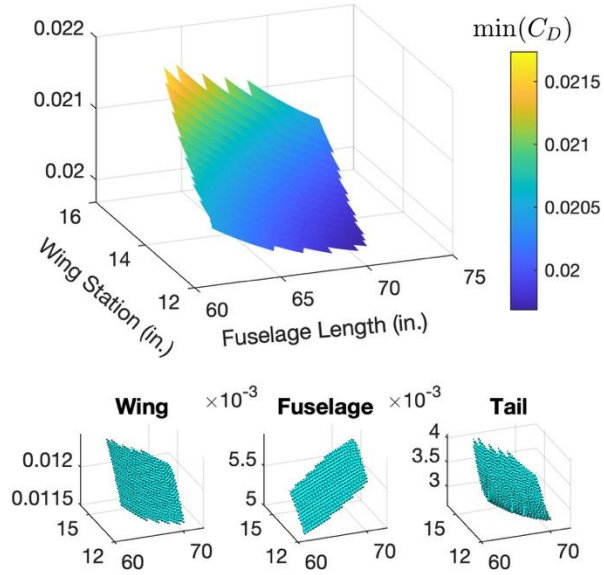


Figure 4.14: Layout MDO Drag Tradeoff Study

The results of the data surveys and subsystem-level MDO analysis were discussed with the chief engineer and experienced former team pilots to produce a preliminary aircraft layout that satisfied both quantitative and intuitive assessments of aerodynamic efficiency, stability/control, structural efficiency and manufacturability. The layout is diagrammed in Figure 4.13, where key components are coded by color.

4.3.4 Stability and Control Verification

The proposed aircraft layout x_3 was assessed *a posteriori* to verify compliance with the higher-fidelity stability constraints. A vortex-lattice surface model of the aircraft was assembled in AVL (see Figure 4.15) to evaluate the aircraft's susceptibility to pecking and Dutch-roll oscillations in M2, arising respectively from short coupling and from low placement of a wing with forward sweep at its quarter-chord. To assess the dynamic effects of internal payload, rotational inertia tensors for both the M1 and M2 configurations were computed using a preliminary weight breakdown. As shown in Table 4.i, static stability derivatives computed at the cruise condition in M1/M2 compared satisfactorily with general-aviation recommendations [13]. As shown in the Root Locus plot in Figure 4.16, the fundamental dynamic modes of the aircraft in both cruise and landing trim, with and without maximum payload, satisfied Level-2 MIL-F-8785C standards (adequate handling characteristics requiring moderate pilot workload) for Class-IV aircraft (highly maneuverable fighter aircraft) in Category-A/C maneuvers (rapid control and precision tracking/terminal flight phases) [14]. Due to the expected timescale of distinct flight segments in the mission circuit and the high potential for wind/gust interference, spiral divergence and Phugoid oscillations were of far less concern than the prospect of pilot-induced pitch oscillations, especially during the landing approaching in M2. The team communicated these results to the test pilot prior to prototype flight testing.

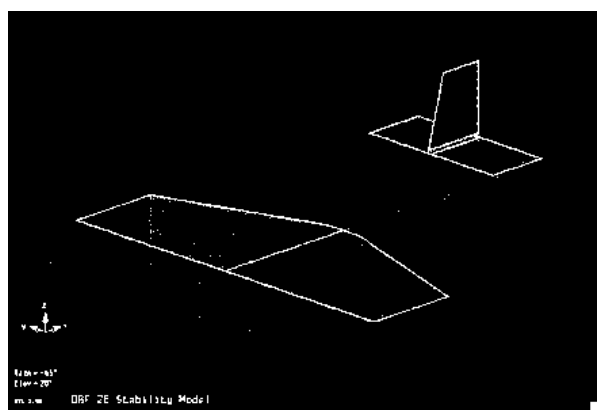


Figure 4.15: AVL Surface Model

Table 4.i: Static Derivatives (rad^{-1}) for Pitch/Yaw Stability/Damping/Control and Dihedral Stability

	$\partial C_M/\partial\alpha$	$\partial C_N/\partial\beta$	$\partial C_M/\partial q$	$\partial C_N/\partial r$	$\partial C_M/\partial\delta_E$	$\partial C_N/\partial\delta_R$	$\partial C_l/\partial\beta$
Standard	–1.5 : –0.3	0.05 : 0.4	–40 : –5	–1 : –0.1	–4 : 0	–0.15 : 0	–0.4 : 0.1
M1 Cruise	–0.75	0.10	–5.51	–0.15	–0.96	0.09	–0.03
M2 Cruise	–0.69	0.10	–5.38	–0.15	–0.95	0.08	–0.03
M3 Cruise	–0.66	0.10	–5.29	–0.16	–0.94	0.08	–0.08

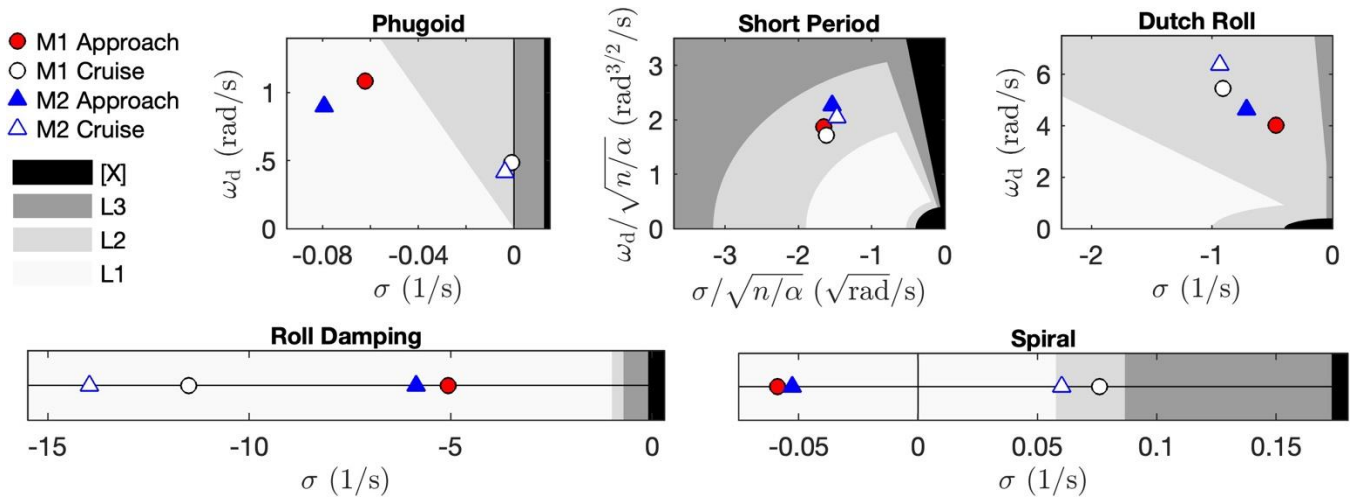


Figure 4.16: Root Locus Plot of Aircraft Dynamic Modes with MIL-SPEC Stability Standards Shaded

In M3, handling characteristics are dominantly influenced by the selection of a proper banner tow point which preserves static stability in cruise and avoids interference between the tow line and tail controls. Given the selected banner stowage method (see Table 3.h), the team considered tow points along the belly and rear fuselage surface to reduce the complexity, size, and installation time of the banner mechanism. Idealistically, the optimal tow point would be that which minimizes the cumulative trim drag energy incurred during an M3 lap due to the resultant moment around the aircraft CG exerted by the banner drag force D_{Bann} and weight W_{Bann}. The magnitude and inclination angle of tow force can be estimated as $(D_{\text{Bann}}^2 + W_{\text{Bann}}^2)^{1/2}$ and $\tan^{-1}(W_{\text{Bann}}/D_{\text{Bann}})$, respectively. Despite ranges of values having been established for both quantities as part of initial sizing, the high uncertainty in flutter-drag prediction as well as the need to test banners with various lengths warranted a robust, versatile tow mount. As suggested in Figure 4.13, the Aerodynamics sub-team recommended the prototyping of an adjustable guide loop through which the tow wire would pass, whose position could be practically optimized using pilot feedback and telemetry measurement of cruise airspeed and energy consumption.

Table 4.j: Stabilizer/Control Surface Geometric Data Survey of DBF 2020 Aircraft

	Horizontal Stabilizer				Vertical Stabilizer				% span : % mean chord		
	AR	λ	Λ_{TE}	t/c	AR	λ	Λ_{TE}	t/c	Elevator	Rudder	Aileron
USC	4.43	1	0	10%	1.92	0.6	0	10%	100 : 37	100 : 36	49 : 26
G-Tech	2.26	1	0	≈10%	1.02	1	0	≈10%	100 : 27	100 : 27	30 : 19
Nevada	2.36	0.76	–	9%	1.26	0.9	–	9%	100 : 60	100 : 58	23 : 20
AARON	2.5	1	0	8%	1.29	0.7	0	8%	100 : 40	100 : 43	40 : 40

Based on the tow-point selection and aircraft dynamic mode analyses, the team deduced that the towed banner would be largely favorable for the stability of the aircraft but require a sufficiently large rudder, to maneuver effectively in crosswinds and combat the accelerated spiral divergence, and a large elevator, to combat the strong

downward pitching moment during competitive high-AoA turns. Thus, all control surfaces were sized based on geometric surveys of successful aircraft specifically in the 2020 DBF competition [2], as enumerated in Table 4.j. The shapes of the tail feathers were also extrapolated from these surveys, favoring gentle sweep angles, small aspect ratios, and taper ratios near unity in order to reduce the likelihood of a deep stall or rudder lock amidst Kansas gusts. Static control derivatives for pitch and yaw compared favorably with general aviation standards, as shown in Table 4.i. The Aerodynamics sub-team also recommended dimensions for optional flaps that would alleviate pilot workload in landing approaches and couple with the ailerons to form large airbrakes to prevent score-nullifying ground-loops after touchdown. Hinge moments for all control surfaces under their limit loads were computed using viscous vortex-panel simulations in XFLR5. As shown in Table 4.k, the Propulsion sub-team employed appropriate FoS to select reliable servo actuators and a battery eliminator circuit (BEC).

Table 4.k: Hinge Moments and Servo/BEC Selection

Control	Limit Load Case	Hinge Moment	FoS	Servo
Rudder	M2 Cruise	0.131 lb•ft	25 %	FrSky Act M5213H
Aileron	M2 Turn Entry	0.387 lb•ft	25 %	
Flap	M2 Approach	0.204 lb•ft	25 %	
Elevator	M2 Dive Pull-up (Aggressive)	0.525 lb•ft	50 %	Hitec HS-645MG
Battery Eliminator Circuit		HobbyWing 3A UBEC (at 6V)		

Due to the high level of epistemic uncertainty in the aerodynamic simulations – especially with respect to the yaw-destabilization effect of fuselage side area – the team constructed and tested two aerodynamic prototypes which successfully validated the selection of tail/control surface sizes, servos, and the banner tow point. Details of the testing are provided in §7 and §8.

4.3.5 Aircraft Drag Breakdown

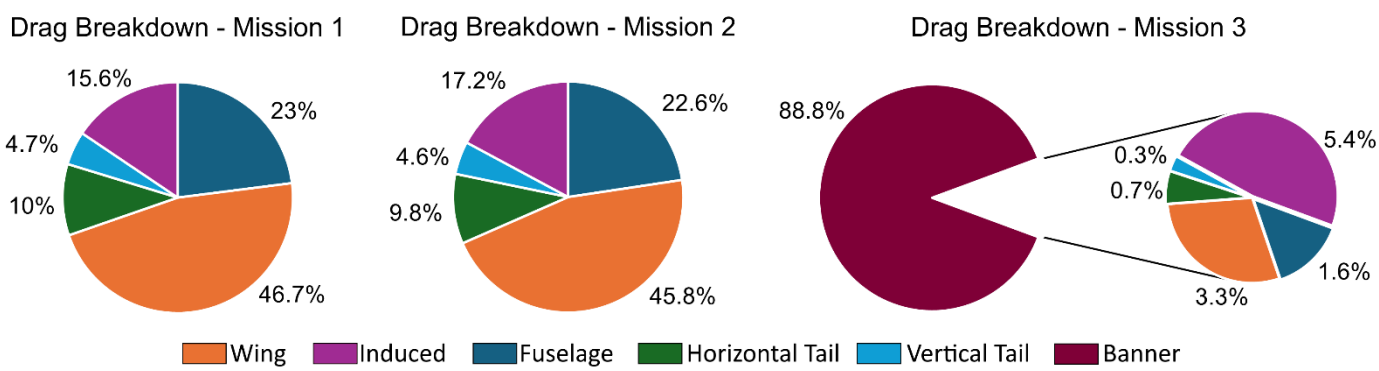


Figure 4.17: Drag Breakdown for M1/M2/M3 at the Cruising Trim Condition

Upon finalization of the OML, a comprehensive drag breakdown was performed using the OpenVSP software, employing three steps. Viscous boundary-element simulations were conducted using the 'VSPAERO' tool to compute the induced drag experienced by the aircraft at the cruising trim condition for each mission. Empirical form-factor calculations were conducted within the 'Parasitic Drag' tool to predict the parasitic drag of each external component on the aircraft at their respective cruising Reynolds numbers [5]. For M3, the team predicted the time-averaged drag force on the flutter-prone banner using an empirical formula assembled using data from the 2020 DBF competition (see Figure 4.3b). The resulting component-wise drag breakdown for all three flight missions is illustrated in Figure 4.17. Notably, the banner drag is expected to dominate all other sources of drag in M3, owing to the large turbulent wake it generates. Induced drag is an increased factor as well, due to a high AoA

required to support the combined weight of the aircraft and banner at the low cruise speed.

Following finalization of the preliminary weight breakdown and OML, improved predictions for state variables $\boldsymbol{y}_3 \subset \boldsymbol{y}$ representing gross weights, and lift and drag polars under the trim conditions for climb-out/cruise/turns, were incorporated into the system-level mission model. Refined values for the maneuver-specific performance targets $\boldsymbol{x}_f$ were found and communicated to the Propulsion sub-team for the selection of a suitable propulsion package.

4.3.6 Banner Design Study

Based on the system-level optimization and the subsystem-level drag breakdown study, the team recognized that performance in M3 will be limited largely by the excess drag force and control interference introduced by the towed banner, as opposed to the incremental contribution to induced drag from its weight. Thus, the team prioritized (i) reduction of unsteady and time-averaged drag and (ii) compliance with rules, in the process of selecting the banner material, banner tow method, and lettering technique. Key factors influencing the banner material selection are summarized in Table 4.l.

Table 4.l: Banner Material Selection Factors

Factor	Consideratios (Pros/Cons)
Areal Mass Density	• Increases critical speed for aeroelastic divergence • Augments flutter amplitude and 'snapping' forces
Durability	• Allows for testing and validation prior to fly-off • Prevents tearing in winds and upon ground impact
Bending Stiffness	• Increases critical speed for aeroelastic divergence • Reduces structural deflection that augments flutter • Improves legibility of lettering in flight • May increase difficulty of stowage during GM
Tensile Elasticity	• May increase effective wetted surface area • Widens pores, potentially increasing skin friction • May distort lettering and cause paint to flake
Porosity	• Triggers early transition to turbulence, inducing excess skin friction while reducing flutter due to flow-induced tension and pressure equilibration • Mesh-type banners ruled illegal in Q&A

Based on extensive literature review surrounding the fluid–structure dynamics of towed banners, the team explored five methods for reducing drag due to flutter, as outlined in Table 4.m. The methods included artificial tensioning and stiffening, and cavities for pressure relaxation. A crucial tradeoff governing drag reduction was identified. Early experiments in the 20th century indicate that surface porosity induces a major drag reduction for fluttering banners, which may otherwise produce a time-averaged drag force that is an order of magnitude larger than the equivalent skin friction force on a hydraulically smooth surface of equal area [15]. In principle, such a reduction in drag owes to the suppression of flutter due to enhanced (i) lateral pressure relaxation and (i) banner tensioning by skin friction [16,17] which is augmented by separation-induced turbulence. However, early test flights using non-porous and porous banners demonstrated that the former induced a nearly 50% relative reduction in endurance, most possibly due to the significant increase in turbulent skin friction. This disparity of results was attributed to a significant difference in the relevant Reynolds number Re_{Bann} between early experiments and the present application. As shown in Figure 4.3b, the relative contribution to drag from flutter reduces by several orders of magnitude as Re_{Bann} increases from the pre-transitional range (< 1e5) toward the values found in the system-level analysis (of order 1e7). In contrast to porosity and to relief cavities used for static

Table 4.m: Methods for Flutter Reduction and Legibility Improvement

Method	Legality
Rear-mounted Parachute	X
Rear-mounted Streamers	X
Artificial Stiffeners (Sewn Fabric / Battens)	X
Wind Relief Doors	√
Longitudinal Wind Slits	√

display banners, longitudinal wind slits have been found to be an effective solution across a wide range of Re_{Bann}. Experiments in the 1930s [18] indicate a 33% drag reduction, while the winning report in the 2020 DBF competition indicates a 15% upon the analogous installation of trailing-edge streamers [2]. Based on these results, the team chose to pursue drag reduction through experimental optimization of a longitudinal slit pattern. The task involves determining the number, positions, and lengths of slits that maximizes endurance at the desired cruise speed while maintaining legibility of the logo and preventing tearing and fraying (as per the rules).

The team considered five lightweight materials for the banner: low-density polyethylene (LDPE), light plastic *Party* tablecloth, and three nylon fabrics: 'sheer stretch' fine mesh, coarse mesh, and 'ripstop'. LDPE and ripstop were given particular attention due to their use by top teams in the similar 2020 DBF competition [2]. Since the updated rules disallowed mesh banners and required a "fabric" material, ultralight 7-denier ripstop was ultimately selected. Upon freezing the banner material, the team surveyed various methods for adding the university logo, maintaining a vertical orientation in flight, and configuring the tow structure. Due to the excessive distributed weight, skin friction drag, and manufacturing complexity introduced by the sewing method, the team opted to utilize fabric-specific spray paint that is resistant to flaking due to substrate bending. Due to the negligible weight involved, concentrated ballast was chosen as the method to preserve verticality. Minimal values for the ballast–banner weight fraction were determined through early prototype testing. Nylon fishing wire was selected as the tow line due to its low density and the wide variety of options available for its tensile strength rating. A spring may be installed along the tow line to suppress the adverse effects of banner 'snapping' dynamics [19] on aircraft control.

4.4 Propulsion System Selection (SSO $i = 4$)

The final subsystem-level optimization problem involved determining the propulsion system that maximizes

$$f_4(\boldsymbol{x}_4) = \sum_{j=2,3} \left[\boldsymbol{k}^{(j)} \bullet \boldsymbol{y}_4\right]_{Mj}, \text{ with respect to } \boldsymbol{x}_4, \text{ subject to } \boldsymbol{c}_4(\boldsymbol{x}_4) \leq 0. \tag{4.4}$$

The set of discrete design variables $\boldsymbol{x}_4$ represents a complete propulsion package, including a battery, an electronic speed controller (ESC), a motor, and propellers for M1/M2/M3. The selection space was constrained by imposing acceptable boundaries for relevant state variables $\boldsymbol{y}_4 \subset \boldsymbol{y}$, including the static thrust-to-weight ratio T_o/W_o, maximum continuous power P_c, and the top speed V_{CR}, endurance $t_{\max}$ and propulsive efficiency η for cruising (in both M2 and M3). The local objective function f_4 was defined as a linear combination of the same state variables, using weightage coefficients $\boldsymbol{k}$ that reflected the degree of importance of each variable to the penultimate objective in each mission, $\mathcal{N}_L$. The functions $\boldsymbol{c}_4$ and coefficients $\boldsymbol{k}$ were deduced using a combination of pilot experience, decomposition of mission requirements (see Table 3.a), and a system-level propulsion constraints analysis (see, for example, Figure 4.8). Due to the absence of a takeoff distance limit and time-based scoring metrics in the 2026 DBF rules, as well as the overwhelming importance of cruising efficiency, the variables V_{CR} and $t_{\max}$ were assigned the highest weights. Quantities such as $[T_o/W_o, P_c, V_{CR}, t_{\max}, \eta]$ were used primarily to rank feasible candidates based on excess power – which is practically crucial for aggressive turns, stall recovery, robustness to headwind, and rapid takeoff/climb-out amidst dangerous gusts.

The MotoCalc and eCalc software were used to expedite the search for suitable propulsion systems that provide the required endurance in M2/M3 while exhibiting robustness to the epistemic uncertainties in the estimates for flying weights (see Table 4.e). In particular, the MotoWizard interface allowed rapid simulation of thousands of

systems to simultaneously determine (i) the propeller geometry (diameter, pitch, and make) that yields the least mechanical power to produce the required dynamic thrust at the target cruise speeds, (ii) the motor size (diameter and KV-rating) that yields the required torque and rotational speed while operating at its state of maximum electromechanical efficiency, and (iii) the battery (chemistry, nominal voltage, and discharge rating) that supplies the required potential difference and current without exhibiting voltage sag. Based on prior team experience and historical data surveys, the search was limited to (i) brushless direct-current motors, due to their superior efficiency, durability and power-to-weight ratio, and (ii) lithium-ion polymer (LiPo) batteries, due to their superior energy/power densities, discharge current, and voltage stability under load. The minimum battery size considered was 5S (five-cell) 5400 mAh (milliamp-hour), to ensure that the current draw in the most demanding maneuver identified in the system-level solution remains below the limit of 100 A (amps) corresponding to the largest fuse allowed by the rules. Upon consultation with prior team pilots and the logistics sub-team, the propulsion sub-team further filtered the set of candidates based on manufacturer repute, availability and item/shipping costs. The results of the selection process are displayed using an abridged selection matrix in Table 4.n.

Table 4.n: Abridged Motor Selection Matrix Based on eCalc/MotoCalc Simulation

Motor	Cost	Mass	Power P_c (W)	Mission 2				Mission 3			
				T_o/W_o	V_{CR} (ft/s)	$t_{\max}$ (s)	η (%)	T_o/W_o	V_{CR} (ft/s)	$t_{\max}$ (s)	η (%)
Hacker A50-10L 530KV	$258	507 g	≈1800	1.17	71	5.9	50.9	1	36	7.4	46.1
Hacker A60-5S V2	$242	595 g	≈2000	0.6	68	9	63.4	1.1	30	11.9	54.1
Scorpion HK-5020-520KV	$370	498 g	4000	0.93	80	5.5	50.2	1.99	42	6.8	47.3
Scorpion HKII-4235-520KV	$400	519 g	4450	0.94	80	5.4	49.6	1.92	42	7.2	56.6
Scorpion SII-4025-520KV	$210	353 g	2000	1.02	85	5.4	50.6	1.96	42	6.9	47.8

Table 4.o: Propulsion System Details

Item	Product	Mass
Motor	Scorpion SII-4025-520KV	353 g
Propeller	M1,2) APC 14 x 8 M3) APC 15 x 8	37 g 44 g
Speed Controller	HobbyWing Skywalker 100A V2	92 g
Fuse	Stinger 100A Maxi Style	–
Battery	Gens Ace G-Tech 6S 4500mAh 60C	658 g ± 10 g

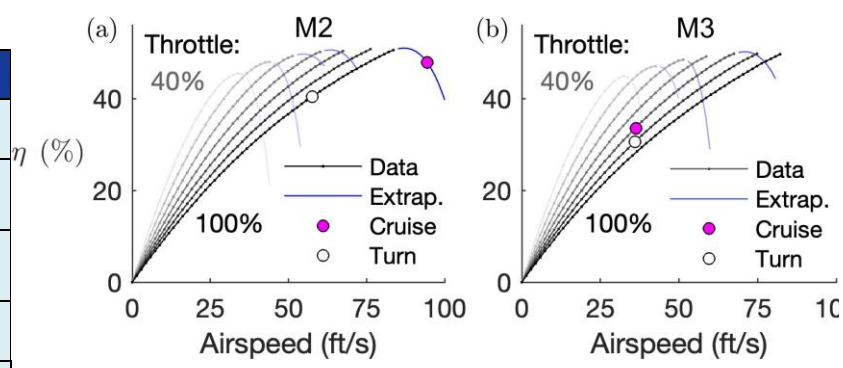


Figure 4.18: Numerically Extrapolated Data for Net Propulsive Efficiency as a Function of Airspeed in (a) M2 and (b) M3

Based on its weight, cost, and performance $\boldsymbol{y}_4$, the Scorpion SII-4025-520KV was chosen as the most appropriate motor. Its performance is compared in Figure 4.18 with that of competing candidates. Based on the current rating of the selected propulsion system, a speed controller and blade-style fuse rated for 100A blade-style were selected. To provide the high continuous discharge current required for M3, a 6S LiPo battery rated for

60C was chosen. The complete propulsion package is described in Table 4.o.

Empirical datasets necessary for the computation of state variables $\boldsymbol{y}_4 \subset \boldsymbol{y}$ representing propulsion system weight, dynamic thrust, battery voltage sag, and electrical power draw were incorporated into the system-level mission model for the final stage of refinement.

4.5 Performance Predictions

Following the final stage of refinement, the system-level optimizer was deployed to produce precise sizing and performance targets for *AARON*, which are listed in Table 4.p. Integration of empirical data characterizing the propulsion system enabled substantially refined prediction of the (i) turning speed and radius, (ii) maneuver-specific power draw, and (ii) net endurance. The expected total mission score for *AARON* is $TMS \simeq 5.4$.

Table 4.p: Performance Predictions for *AARON* Produced Using *RRDO* Distributed Optimization Framework

Parameter		M1	M2	M3	GM
All-Up Weight (lb)		9.05	15.05	10.23	—
Payloads	**Quantity**	None	38 Passengers, 12 Cargo	30.5 ft-long Banner	—
	Weight (lb)	0	6.16	1.23	
Cruise	**Speed (ft/s)**	65	94.5	36.3	—
	Throttle (%)	68.9	100	80.6	
	Current (A)	9.0	22.8	45.3	
Turns	**Speed (ft/s)**	60	57.6	36.0	—
	Radius (ft)	50	43.8	32.1	
	Throttle (%)	79.0	100	91.5	
	Current (A)	25.5	62.0	63.9	
Battery Drainage (%)		11	55	79	—
Number of Laps		3	8	4	—
Mission Time (sec)		131.8	252.2	268.0	133
Mission Score		1	0.53	0.64	0.23

4.5.1 Uncertainty Quantification

The team recognized that there is significant room for improving the robustness of the *RRDO* architecture to epistemic uncertainties surrounding the subsystem-level prediction of aerodynamic, electromechanical and structural efficiencies. Thus, the team is in communication with their faculty advisor to assemble an integrated stochastic optimization framework where (i) data surveys are used to define appropriate moments for all input parameters of interest, (ii) the error propagation through system- and subsystem-level analyses is rigorously quantified, and (iii) the optimization algorithm prioritizes designs that maximize the scoring objective while also minimizing the level of uncertainty due to its dependence on parameters with wider uncertainty intervals. Historical data surveys [2] and telemetry data collected during upcoming flight tests will be used extensively in developing a rigorous methodology for defining the moments of interest.

5. Detailed Design

The results of the preliminary design process described above provided a set of guidelines for the detailed design phase. This phase revolved around transforming the theoretical design into a feasible prototype to be built and tested. Balancing structural strength with mass properties are the guiding principles of the detailed design.

5.1 Aircraft Dimensions

Table 5.a: Finalized design parameters

Wing		Fuselage	
Longitudinal Location (in)	21	Total Length (in)	66
Span (in)	60	Width (in)	6
MAC (in)	20.35	Average Height (in)	8
Leading Edge Sweep (°)	0	Fineness Ratio	9.5
Taper Ratio	0.625	Wetted area (in^2)	798.24
Planform Area (in^2)	1197	**Balance (MTOW: M2)**	
Aspect Ratio	3	CG Location (in)	28.9
Aileron Span (in)	12 (40% span)	Neutral Point Location (in)	25
Aileron Average Chord (in)	3.4 (20% chord)	Design Static Margin	19.1
Vertical Tail		**Horizontal Tail**	
Airfoil	NACA 0008	Airfoil	NACA 0008
Span (in)	11	Span (in)	25
Root Chord (in)	10	Root Chord (in)	10
Tip Chord (in)	7	Leading Edge Sweep (°)	0
Planform Area (in^2)	93.5	Planform Area (in^2)	250
Rudder Span (in)	11 (100% span)	Elevator Span	25 (100% span)
Rudder Chord (in)	3.5 (41% chord)	Elevator Chord	4 (40% chord)
Tail Volume Coefficient	0.15	Tail Volume Coefficient	0.4

5.2 Structural Design and Analysis

The structures team began the structural design by taking expected flight loads from the aerodynamics team in the preliminary design along with chosen load-factors. These constraints, shown in Figure 5.1, allowed for the aircraft structure to be sized component by component. A balsa and basswood build-up was done for all structures except the tail, which was a foam-core composite wet layup.

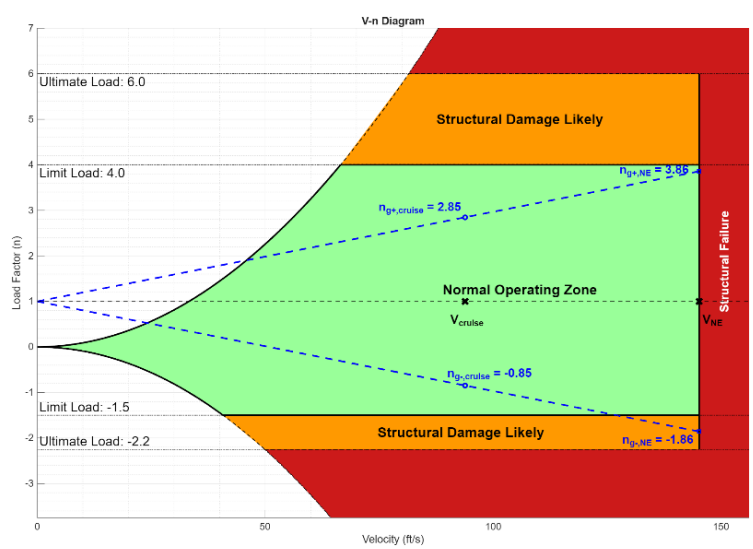


Figure 5.1: V-n Diagram

5.2.1 Wing Structure

The team decided on COTS composite tubes as the primary load bearing structure. Initial sizing estimates for the spars were used with an analytical Euler-Bernoulli solver, which assumed a spanwise lift distribution in the form of a fourth-order polynomial and evaluated stresses at four points in the spar cross section. When scaled with the design MTOW and maneuvering load factor n = 6, the lowest margin of safety was found to be 0.24, using $\sigma_{allow} = \frac{\sigma_{yield}}{SF}$ and a safety factor of 1.1. An ABAQUS FEA simulation of the composite structure was later run with a higher fidelity lift distribution provided by XFLR5 under the same flight conditions and determined center of pressure, which confirmed the preliminary analysis. Tabulated stress values and yield strengths can be found in Table 5.b. The remaining internal wing structure consists of wooden ribs, stringers, and spar near the trailing edge separating the wing from its control surfaces. While the main carbon spar carries the primary bending loads, a secondary carbon fiber rear spar helps with the attachment and load distribution of the main landing gear system. The overall wing structure has a total half span of 29.75 in. with a root chord of 24 in. and a tip chord of 15 in., yielding a taper ratio (λ) of 0.625. To minimize mass, the wing is primarily made from 1/8 in. balsa wood, with the exception of the wing box attached to the wing and the leading-edge closeout. The wing box is made with purely bass wood to increase stiffness at the fuselage attachment point, whereas the leading-edge closeout consists of 1/32 in. balsa sheeting. The outer surface of the wing was then covered in Monokote film.

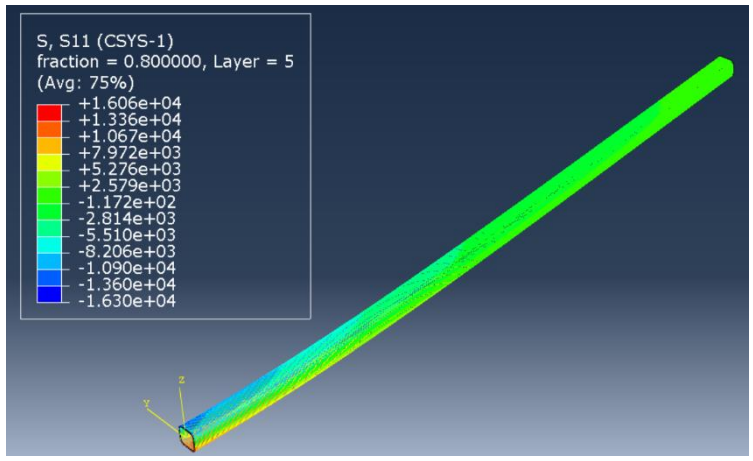


Figure 5.2: ABAQUS FEA Simulation of Wing Spar

Table 5.b: Wing Spar Structural Analysis

Max Stresses	Analytical Approximation	ABAQUS Solution	Allowable Stress	Unit
S11 – Axial Stress, Tension	15.65	16.06	45.87	ksi
S12 – Shear Stress XY	1.34243	1.254	8.667	ksi
S23 – Shear Stress XZ	6.1223	5.106	8.667	ksi
Margin of Safety	1.90	1.86	-	1

5.2.2 Fuselage Structure

A primary structural engineering challenge for *AARON* was maximizing usable internal fuselage volume while satisfying requirement RQ-M2.2. To address this, *AARON*'s internal structural layout consists of an array of "H"-shaped bulkheads extending from the firewall to the base of the empennage. These bulkheads function as shear frames, creating large internal cavities for

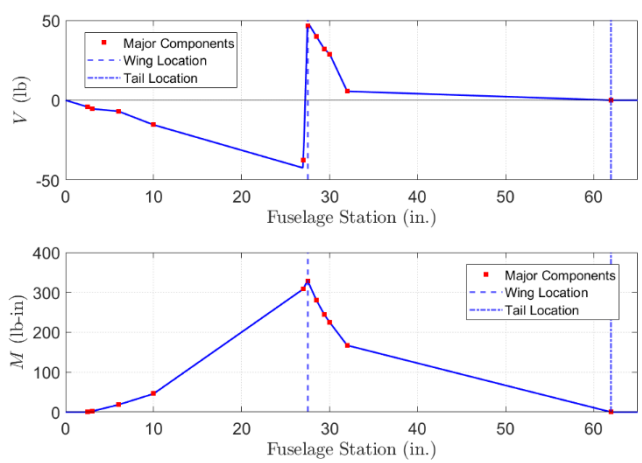


Figure 5.3: SFD and BMD for Fuselage

passenger loading and cargo placement while stabilizing the balsa fuselage skins against shear deformation, buckling, and panel distortion. Additionally, the bulkheads support integrated passenger and cargo floorboards spanning between fuselage walls. Beyond satisfying the competition requirement to separate passengers from cargo, the passenger and cargo floorboards work in conjunction with the bulkheads to improve longitudinal bending resistance, reduce shear deformation between parallel fuselage walls, and increase the overall torsional rigidity of the fuselage structure. *AARON*'s mid-fuselage bulkheads extend below the fuselage centerline to form the wing box structure, housing both the primary and aft wing spars. This design configuration provides a very robust attachment point between the wing and fuselage, enabling the fuselage structure to react to the bending and torsional loads created by the lower wing configuration. Attention was given to ensuring that the lift-induced loads generated from the wing are effectively transferred through the spars, into the connected bulkhead frames, and then distributed into the fuselages' longer structural members. Overall, this bulkhead frame-to-wing box system creates a semi-monocoque structure that maximizes allowable internal passenger and payload volume while preserving structural stiffness, load carrying capabilities, and aerodynamic conformity across the expected flight envelope of operation. Component-wise bending moment and shear force diagrams were created to analyze the expected fuselage loading during maneuvering, as seen in Figure 5.3.

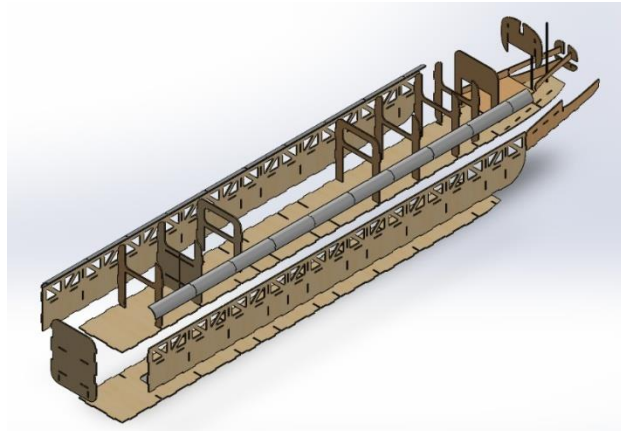


Figure 5.4: Fuselage Structure Exploded View

5.2.3 Tail Structure

During tail design, load paths such as bending due to tail downforce, twisting during turns, and longitudinal bending were considered. A carbon fiber composite with a foam core was selected as the material for the vertical and horizontal stabilizers, attached to the basswood structure of the fuselage. The walls leading to the tail structure were kicked aft from the passenger side walls towards the tip of the fuselage to counteract the load paths described above. An additional horizontal structure was added to increase the moment of inertia and thus prevent longitudinal bending. The vertical stabilizer structure is attached using two carbon fiber rods running spanwise into the vertical stabilizer mounted to intersect both the horizontal stabilizer and the passenger floor to prevent any translation of the tail structure.

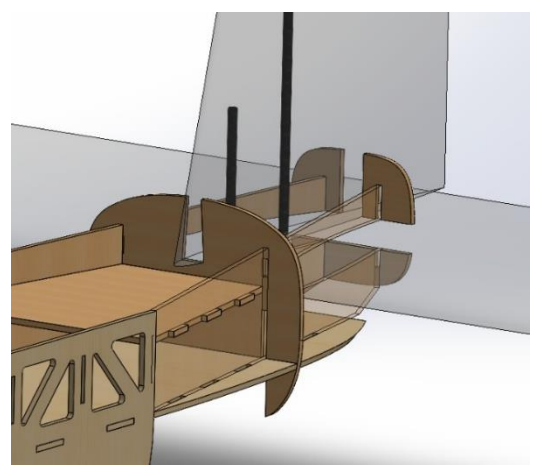


Figure 5.5: Tail Structure Build-up

5.2.4 Passenger Compartment and Restraint

The fuselage interior is divided into upper and lower compartments, separated by a horizontally oriented floorboard. Passengers are seated above this floorboard while the cargo is placed below in a configuration conceptually similar to that of a traditional commercial airliner. *AARON*'s internal fuselage layout was specifically designed to provide sufficient usable volume to accommodate 38 rubber duck passengers of varying shapes and sizes while maintaining structural efficiency. The floorboard is mounted perpendicular to the fuselage sidewalls via rectangular slot insertions, creating a structurally integrated platform that assists in separating payload compartments while also contributing to fuselage stiffness and load distribution. After careful consideration, *AARON*'s passenger restraint system was designed as a patterned array of dome-shaped enclosures that secure the passengers during flight. Dimensional data (height, width, and length) were collected for all duck models, and the restraint geometry was sized to accommodate the largest dimensions observed in each category. A standardized dome configuration allowed the ground crew to organize the passengers efficiently in a two-row seating arrangement without concern for fit variability. The overall design objective is to streamline the passenger onboarding process while maintaining consistent restraint performance and minimizing loading time during mission operations.

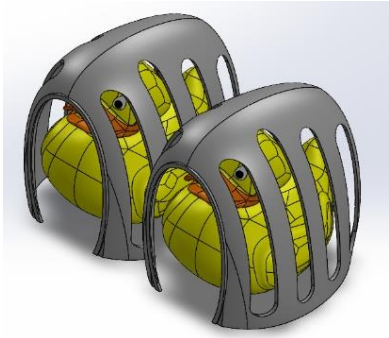


Figure 5.6: Passenger Restraint

5.2.5 Passenger Access Hatches

Direct access to *AARON*'s passenger compartment for boarding and disembarking was provided through three individual hatches located along the top of the fuselage. Inspired by the clip-lock mechanisms commonly used in electronic battery compartments (e.g., EP-0926-023-9V battery box [20]), a similar clip-lock design was developed to securely retain the hatches under expected flight loads and maneuver conditions. The selected mechanism provides a balance between secure retention and operational accessibility, satisfying requirement RQ-M1.1 and RQ-GM.2. This configuration enables efficient access to the three passenger sections while maintaining structural integrity and minimizing the risk of the top hatches inadvertently opening during mission flights.

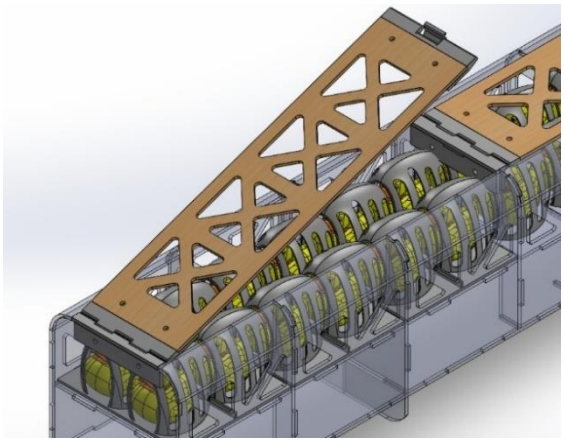


Figure 5.7: Hatch Mechanism

5.2.6 Cargo Compartment and Access

The design team needed to find a way to efficiently load the payload into the plane and be able to displace it as close to the calculated CG as possible to allow the plane its desired flight maneuverability. The cargo portion of the payload consisted of a balsa box, with 12 hockey pucks in a two-puck high and six-puck length configuration. For the cargo compartment access, a door beneath the empennage of the plane opens and allows the cargo to slide in below the passenger compartment, similar to a C-130 Hercules. Once the cargo box is shifted to the

desired location within the fuse, it will then be locked in place with the use of a CAM lock mechanism. This will use a single CAM lock shaft that can be slid through side ports of the fuselage, preventing the cargo from displacing through the cargo compartment cavity as *AARON*. performs mission-required maneuvers.

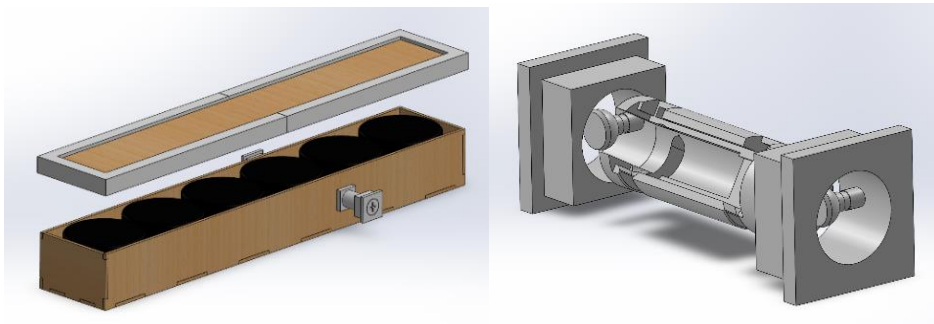


Figure 5.8: Cargo Compartment and Fastener

5.2.7 Banner Release Mechanism

For M3, the banner will be rolled up and stowed underneath the fuselage. It was then held in place and then deployed by banner deployment mechanisms and then towed and released by the banner release mechanism. To comply with GM requirements without compromising M1 and M2 performance, the banner deployment and release system will be part of a slide mount assembly that is detachable from a permanent mount below the fuselage. The slide mount consisted of two banner deployment mechanisms integrated into the mount. Each deployment mechanism is a 3-bar linkage with a guide loop, made of fishing wire. A metal ring on the end of the guide loop will be pulled around the banner and secured on the servo pin within the u-block. When the servo arm is rotated, the pin is removed from the u-block, and the bands are released, allowing the banner to deploy. For the banner release, a simple servo pin lock mechanism tows and releases the banner after the final lap.

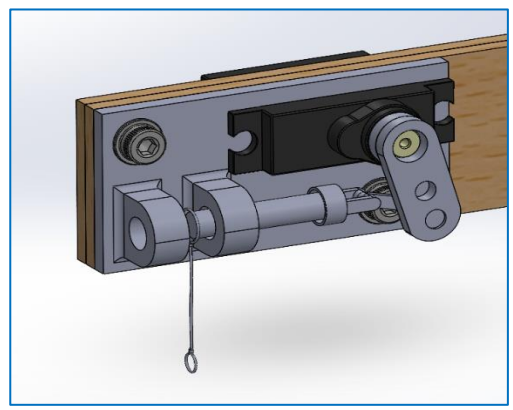


Figure 5.9: 3-bar Linkage with Guide Loop

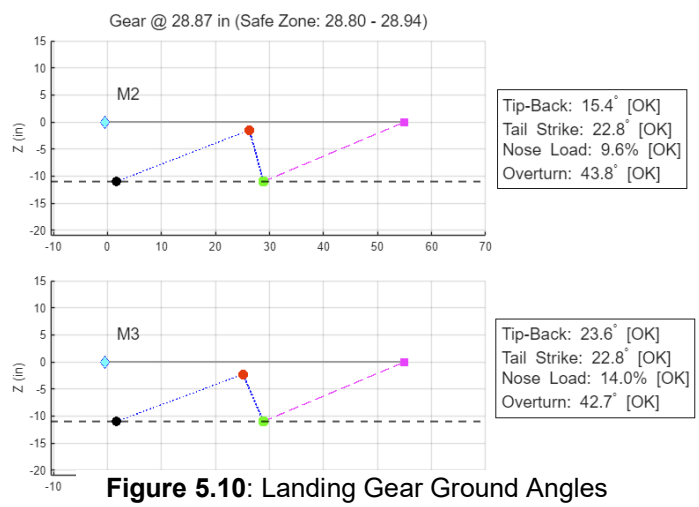


Figure 5.10: Landing Gear Ground Angles

5.2.8 Landing Gear Design and Analysis

As summarized in the Pugh Chart for Landing Gear Configuration (Table 3.k) in §3, the team selected a retractable tricycle gear configuration. The integration of the gear relies on a custom-designed modular mounting box that ensures structural continuity. This modular assembly, similar in concept to the main wing box, is mechanically tied to both the forward and aft wing spars. This dual-spar connection effectively transfers bending moments and lateral ground loads into the primary wing structure. Geometrically, the units are oriented to retract

inboard toward the fuselage. A 3D-printed fairing is installed atop the mounting box to create a flush surface profile, ensuring a seamless application of the Monokote covering the wing skin.

A detailed mass breakdown was used to compute the Center of Gravity (CG) for three mission scenarios (M1: no payload, M2: full payload, M3: banner), using standard mass-weighted averages. The geometric constraints used in the code align with standard initial sizing criteria for general aviation aircraft and UAVs (for example, Raymer). A 15-degree tip-back angle provides a margin against bumps, braking pitch, and rotation variability. A tail strike angle of less than 15 degrees provides a safety buffer for takeoff rotation. Nose gear load (< 15%): ensures effective steering traction without exceeding the structural capacity of the nose strut. Overturn angle (< 63 degrees): verifies that the track width is sufficient to prevent rolling over during a turn. FEA of MLG was performed using the worst-case scenario of a load of 6Gs (90 lb) sustained on a singular LG system in landing configuration. For this preliminary analysis, the rubber tire was removed due to nonlinearity. The maximum von-Mises stress was found to occur at the connection between the axle and the wheel at 74 ksi. With 7075 T6 aluminum was used for the MLG strut and a yield strength of 74 ksi, the analysis resulted in a FoS of 1. However, as suspension and small features like fillets and bearings in that area were not modeled, accounting for these factors will raise the FoS sufficiently.

Figure 5.11: FEA Simulation of Main Landing Gear

5.3 Avionics and Wiring

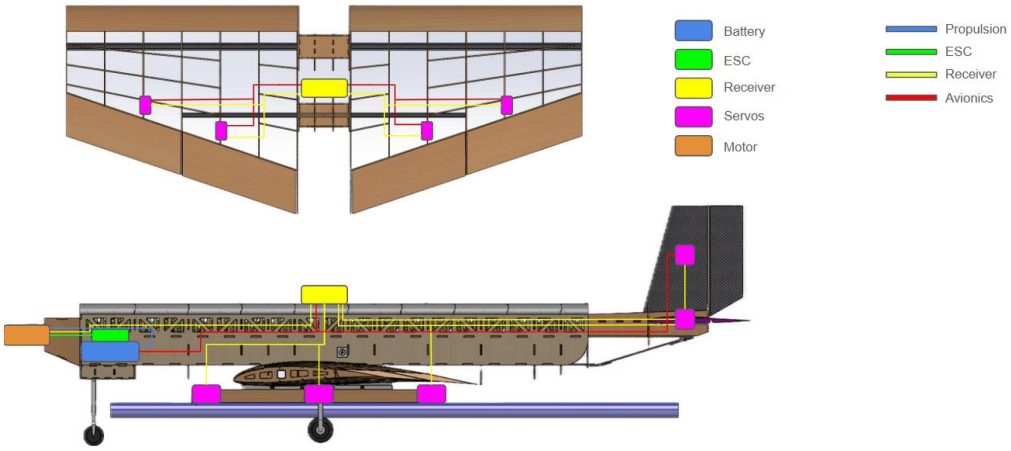


Figure 5.12: *AARON* Wiring Diagram

The team focused on a modular wiring system to comply with critical load paths and necessary internal structure. The aircraft runs on a mono-propellor tractor configuration where a SII-4025-520KV motor was connected to a 100A ESC, 100A fuse, and a 6S 4500mAh battery. The battery also powers the servos required for the banner mechanism and control surfaces. A switch was integrated to disconnect the receiver from the battery if necessary.

Servos were sized based on the predicted torque requirements from Table 4.k.

5.4 Weight and Balance

Weights and positions for all components for *AARON* were measured using scales and are listed in Table 5.c. Certain components that were coupled with surrounding structure were grouped into one. Positions of components were found using the Solidworks mass properties function. Weights for certain missions were calculated and are comparable to the predicted weights in §4.5.

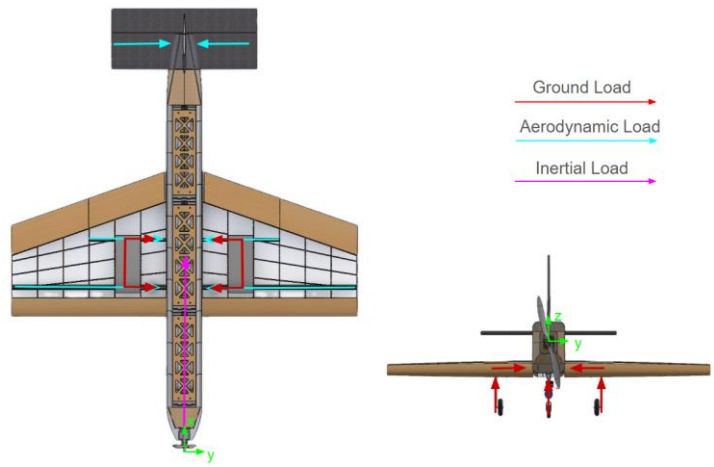


Figure 5.13: Load Path Diagram

Table 5.c: Component-wise and cumulative weights and CG

Component	Mass (lbf)	x-position (in)	z-position (in)
Fuselage	2.31	34.44	-0.32
Wing	1.65	31.65	-3.88
Horizontal Tail	0.25	67.29	1.38
Vertical Tail	0.18	67.26	6.60
Propeller	0.08	0.51	0
Motor	0.78	2.83	0.00
Battery	1.42	10.10	-2.02
ESC	0.20	10.10	-1.5
Motor Mount	0.10	5.16	-0.35
Nose Landing Gear	0.63	8.73	-5.85
Main Landing Gears	1.45	30.71	-5.77
Mission 1 CG	**9.05**	**25.40**	**-2.31**
Passenger Hatches	0.25	31.79	2.83
Cargo Hatch	0.17	61.85	-1.86
Passenger Compartment	2.18	31.92	0.77
Cargo Compartment	3.40	27.08	-1.75
Mission 2 CG	**15.05**	**27.24**	**-1.64**
Banner Mechanism	0.40	35.43	-6.73
Banner	0.78	38.01	-7.44
Mission 3 CG	**10.23**	**26.75**	**-2.87**

5.5 Drawing Package

6 Manufacturing Plan

6.1 Selection of Manufacturing Processes

The team evaluated the diverse manufacturing processes required for each aircraft component and its complexities and effectiveness. The manufacturing processes and purposes of each are described below:

6.1.1. Laser cutting wood

Utilization of laser cutting wood components allows for precise machining of structures that are under substantial amounts of load. The accessibility and low cost of laser cutting balsa wood allow for the fabrication of fuselage bulkheads and wing ribs to maintain a lightweight composition while preventing human errors and allow for faster manufacturing time perfect for prototyping.

6.1.2. 4-Axis CNC Foam Wire Cutting

4-axis CNC wire cutting is useful for carving complex aerodynamic profiles from XPS or EPS foam blocks. This method serves as an efficient means of producing lightweight internal foam cores while maintaining accurate and consistent results.

6.1.3. Composite Manufacturing

Composites are utilized for their high strength-to-weight ratio and as a cost-effective alternative to creating strong components. Composite manufacturing often involves wet-layups, or laying of epoxy infused or pre-impregnated composite fibers, on foam cores and letting it cure in a vacuum bag within a mold to remove excess resin and air bubbles. Specifically, composite layering enables manufacturing complex shapes easily as strong composites, such as fiberglass, carbon fiber, or Kevlar, can be molded to form a desired shape.

6.1.4. Ironed Monokote Plastic Film

With the application of heat using an iron, thin plastic film called Monokote becomes malleable, allowing for the contouring of the balsa wood inner structure to achieve optimal flow over the shape of the aircraft. Once Monokote is cooled, it retains its shape, acting as a lightweight, semi-rigid shell.

6.1.5. 3D Printing

Additive manufacturing, or often known as 3D-printing, offers the ability to fabricate complex geometry tailored to the plane's fuselage at low cost while maintaining functionality and consistency. Specifically, this is used to create small, mission-specific components or similar parts that won't be subjected to high loads, while remaining light weight.

6.1.6. Balsa Skinning

Balsa skinning allowed for thinner but stiffer material to comply with the geometry of the wing's leading edge closeout. By dampening the wood and molding it to the desired geometry, the part would maintain its shape after drying.

6.2 Material Selection

Table 6.a: Manufacturing Selection Process

		Materials				
Characteristics	**Weight**	**3D-Printing**	**Composites**	**MonoKote**	**CNC Cut Foam**	**Balsa**
Weight	5	4	4	5	4	5
Strength	5	3	5	1	2	3
Time	4	4	2	4	4	3
Experience	3	4	1	4	4	4
Cost	2	4	1	3	4	3
	Total	63	58	64	66	70

Material weights and strengths were weighted highest, as they were directly correlated with aircraft mission performance. Turnaround time was next, as the team needed to quickly transition from manufacturing the prototype plane to the competition plane. Experience was then considered before cost. Because the team had limited experience with composites, the competition plane was decided to be a balsa and Monokote build, with custom 3D printed parts for some mechanisms.

Table 6.b: Manufacturing Process Selected for Different Parts

Part	Material	Technique
Wing	Balsa, Bass wood	Laser Cutting
Fuselage		
Empennage	XPS Foam, Carbon Fiber	Composite Manufacturing
Tail Boom and Spars	Carbon Fiber	Purchased (COTS)
Skin	Balsa, Monokote Film	Balsa, Hot Iron
Landing Gear	Aluminum	Purchased (COTs)

6.2.1 Fuselage

The fuselage was built-up with a combination of bass wood, balsa, and Cyanoacrylate (CA). Bass wood was chosen for more critical load-bearing components such as the cargo floor, bulkheads, and side walls that handle both landing and aerodynamic loads. Balsa was used for less critical structures such as the passenger floor and access hatches. All components were laser-cut from 1/8-inch stock, with lightening holes added where appropriate. Both Monokote film and 3D printed fairings were applied externally to maintain the OML geometry. To compromise between high-volume payload and unwanted drag from excess wetted area, the final fuselage dimensions were derived from GM testing data that established optimal passenger and cargo placement.

6.2.2 Main Wing

The main wing was manufactured using balsa, plywood, CA, and a carbon fiber spar. The wing ribs and stringers were laser cut out of 1/8-inch balsa, and CA was then used to bond the ribs and stringers together to stiffen the wing skin to prevent buckling. The main carbon fiber twill woven spar was commercially manufactured with a 0.880” outside diameter and is used to withstand bending induced shear loads. On the leading edge of the wing, balsa skinning was utilized to maintain the aerodynamic surface

Figure 6.1: Construction of Prototype Wing and Wing box

and to prevent the leading edge from buckling compared to using Monokote film. Monokote film was applied across the whole surface of the main wing to maintain its aerodynamic profile. The aileron and flaps were 3D printed in light weight PLA plastic to simplify manufacturing as its complex surface requires a maintained airfoil shape.

6.2.3 Tail

The vertical and horizontal tail were manufactured using composite wet layup including fiberglass, carbon fiber and Kevlar over a foam core. Adequate sheets of fiberglass were utilized to reinforce the structural integrity of the tail. The carbon fiber was used to reinforce the leading edge to sustain aerodynamic loads. Subsequently, Kevlar strips were cut out to reinforce the hinge line for moments induced during maneuvers. Epoxy was then coated over fiberglass sheets with Kevlar attached to the hinge line and carbon fiber strip on the leading edge along the foam core. Finally, the wet layup was vacuum sealed to evenly distribute pressure around the surface ensuring consistent shape, structure, and mitigate air bubbles during curing. Additional carbon fiber rods were inserted spanwise through the vertical stabilizer and intersected with the horizontal stabilizer and fuselage floor to take out the moment induced by winds.

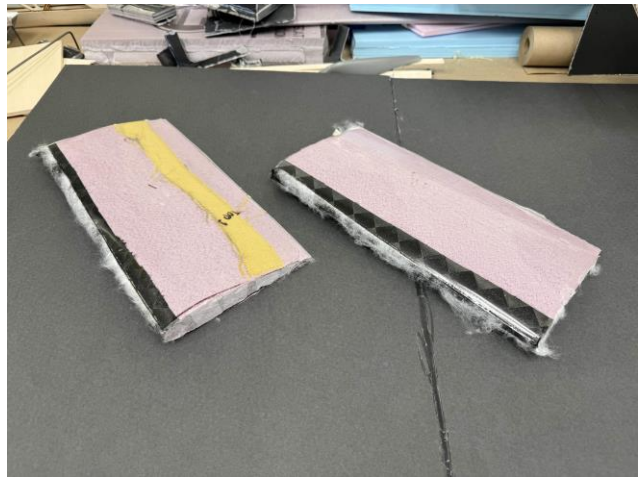


Figure 6.2: Vertical and Horizontal stabilizer Composite Layup

6.2.4 Payload Systems

For components that were not load bearing, different types of filaments were evaluated for 3D printing based on weight and durability. It was decided that lightweight PLA was the most efficient choice, for its low mass density of 540 g/cm^3. The passenger restraint systems and latch mechanisms were 3D printed with this material and coupled with balsa when necessary. The cargo box was made from laser-cut balsa to minimize weight and maximize simplicity.

Figure 6.1: Passenger Restraint Mechanism

6.3 Manufacturing Timeline

This year, with the choice of balsa Monokote construction for the majority of major components, the team benefited significantly from the guidance of alumni advisors and a graduate student mentor/pilot during both the manufacturing process and material selection. These changes, along with the collaborative selection of materials, are detailed in the fabrication timeline and Gantt chart shown in Figure 6.2.

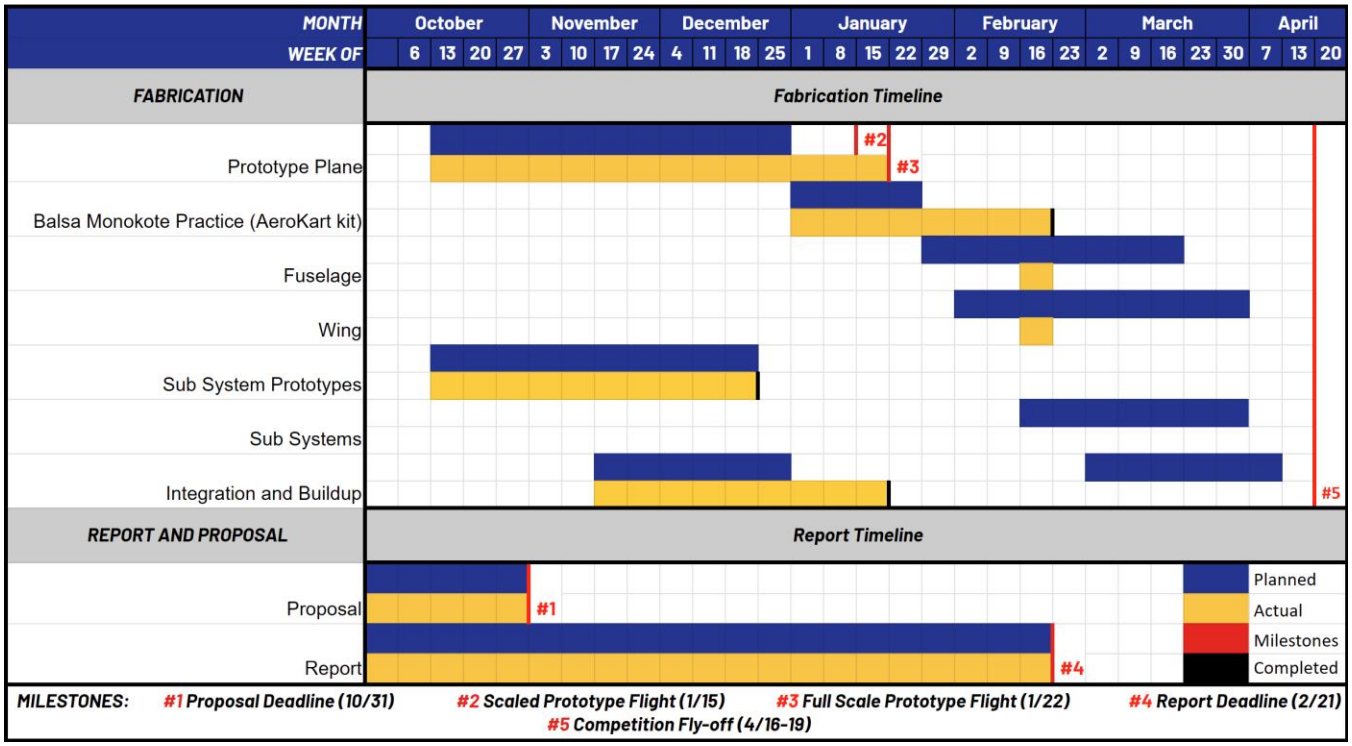


Figure 6.2 A Gantt chart displaying the planned fabrication schedule.

7. Testing Plan

A testing strategy was developed that follows a risk-reduction methodology, progressing from component-level validation to full-system mission simulation. This approach, outlined in Table 7.a, ensures that critical subsystems are verified against analytical models before integrated flight testing begins, minimizing the risk of catastrophic failure during airborne operations.

Table 7.a: Test's Summary

Test name	Main objective	Procedure summary
Wing Load/Deflection	Validate limit-load strength and deflection	Incremental loading to limit load; measure deflection at each step.
Landing Gear Drop	Verify compliance with landing impact loads	Drop from specified height; inspect deformation and rebound.
Fuselage Bending	Confirm bending stiffness and load path	Apply known moment; measure displacement vs. predictions.
Static Thrust Curve	Determine thrust vs. throttle and efficiency	Sweep throttle 0–100%; log thrust, current, voltage, and ESC temp.
Battery Endurance	Validate mission-duration capability	Run simulated mission throttle profile; track voltage sag and temps.
Telemetry Range	Confirm reliable link across flight envelope	Range walk test with RSSI logging; check failsafe triggers.
Servo Torque Margin	Ensure servos exceed hinge-moment needs	Apply known loads; measure stall torque and current draw.
Passenger/Cargo Loading	Measure loading/unloading time & retention	Time trials for loading; retention tests under vibration.
Banner Tow & Release	Validate tow force capability and release reliability	Progressive tow-force test; multiple deployment and release cycles.
Taxi Stability	Evaluate ground handling and steering control	Low and medium speed taxi runs with video capture.
Maiden Flight	Verify stability, control, and stall margins	Short flights; perform trim, basic maneuvers, and controlled landing.
Mission Rehearsal M1	Validate M1 mission profile vs. predictions	Run M1 profile; log lap times, battery use, and throttle.
Mission Rehearsal M2/M3	Validate payload/banner missions vs. predictions	Run M2/M3 profiles; collect mission metrics and deployment data.
Predicted vs Actual	Compare actual test data to models and update	Create comparison tables, analyze deltas, and update models.

The following quantitative goals in Table 7.b have been established to verify that the aircraft meets all mission requirements. Success is defined by meeting or exceeding the "Acceptance Criteria" thresholds derived from the team's performance models.

Table 7.b: Quantitative goals

Domain	Key Performance Metric	Target / Acceptance Criteria	Verification Method
Aerodynamics & Structure	• Wing Tip Deflection • Landing Gear Durability • Stall Characteristics	• **20–30 mm** @ Limit Load • **≤ 2 mm** deformation • **Benign** (no wing drop)	• Static Load Test • Drop Test • Flight Test
Propulsion	• Static Thrust • Thermal Margin (ESC) • Battery Endurance	• **≥ 9.5 lbf** @ 80% Throttle • **≤ 80°C** peak temp • **≥ 8 min** (M2 Profile)	• Thrust Stand • Bench & Flight Log • Simulated Mission Cycle
Mission & Payload	• Loading Time • Telemetry Range • Banner Tow	• **≤ 35 s** (Passenger/Cargo) • **≥ 800 ft** (No Failsafe) • Successful Release (100%)	• Timed Ground Trial • Ground Range Walk • Ground & Flight Test

7.1 Testing Schedule

The testing plan for the 2025-2026 DBF aircraft is designed to evaluate structural integrity, flight performance, and component reliability. By conducting systematic tests, the team aims to identify design optimizations and mitigate potential failures before competition. The schedule for testing the subsystems and aerodynamic performance is defined in Figure 7.1.

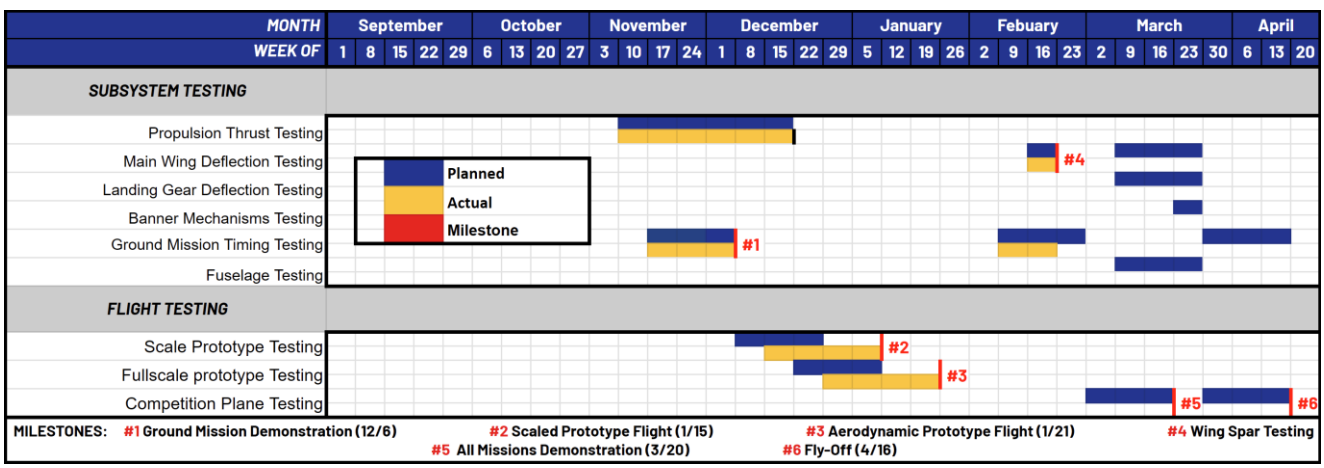


Figure 7.1: Gantt chart detailing the timeline for planned tests

7.2 Sub-system Testing

7.2.1 Payload Systems Testing

The payload systems testing was done early in conceptual design and was iterated upon throughout the detailed design and manufacturing phases. Foamboard mock-ups of the cargo and passenger compartments were made from foamboard and dimensioned to allow for efficient loading and access during ground mission. GM simulations were then conducted, with results informing GM predictions that were implemented in the mission model of the *RRDO* architecture. In conjunction with the passenger compartment, the access hatches

were prototyped and tested iteratively to ensure robustness during ground mission and reliability during flight.

Figure 7.2: Passenger/Cargo System Mockup (left) and GM Prototype V1 (right)

7.2.2 Static Thrust Testing

A static thrust test was conducted to validate power performance and theoretical thrust of propulsion systems. From utilizing a variety of throttling settings ranging from 25-100%, thrust, current, and power draw were examined to determine if experimental data agrees with theoretical predictions. Testing was performed on a Tyto Robotics Series 1585 static test stand equipped with a Scorpion SII-4025-520KV motor and a HobbyWing Skywalker 100A V2 Brushless ESC. Power was supplied by a Gens ace G-Tech 6S 4500mAh 60C 22.2V LiPo battery. To control the motor and log the data, the test setup used Tyto Robotics' RCbenchmark software, which also displayed telemetry live and recorded measurements as a function of time across the throttle sweeps. The collected data was imported into MATLAB for closer analysis. Through iterations of testing multiple propellers, the propeller selection was based on desired theoretical performance results for specific flight profiles. Specifically, a 14x8 APC electric propeller was evaluated for M2 requirements, and a 15x8 APC Electric propeller was tested for the heavier payload requirements of M3.

Figure 7.3: Static Thrust Testing Setup

7.2.3 Structural Testing

A structural test was performed to validate the wing spar's ability to withstand aerodynamic and landing loads. The square composite tube, with outer diameter of 0.88 inches and inner diameter of 0.75 inches, was subjected to a load in a simply supported configuration. Supports were placed 13" from the center line, or 43% of the half span, to simulate a point load introduction at the centroid of the predicted lift distribution from Figure 4.11. For a 6g banked turn, this section was predicted to experience a load of 90 lbf. In the test setup shown in Figure 7.4, two 45 lb weights were placed at the centroid of the predicted lift distribution. The carbon

spar did not buckle or break under the tested load, demonstrating the wing's ability to support the aircraft weight and banked turn loading conditions.

Figure 7.4: Wing Spar Structural Test Setup

7.3 Flight Testing Plan

Repeated flight tests were carried out using various mission configurations to evaluate the performance of metrics and ease of control for the pilot.

7.3.1 Prototype aircraft

In the prototyping phases, two separate flying prototypes were constructed in addition to several subsystem mockups. The 40% scaled prototype, shown below in §7.3.2, proved the flight configuration could fly and tow a banner. The full-scale prototype, shown in Figure 7.5, was constructed using a material downscale but similar structural components and geometry to that of the competition aircraft design. While the prototype maintained the competition aircraft's structure, a spar and rib wing with skinning, and a skinned fuselage with internal H shaped bulkhead, the internal structure was built-up with different materials. In effect, the aerodynamic full-scale prototype was constructed in the same way as the competition plane, but the materials of each component were downscaled, the balsas component on the competition aircraft like the fuselage, wings and control surfaces were built with foamboard on the prototype. The more critical structural components which would be made of stronger materials like carbon fiber composites and bass wood were made with balsa instead. Full-scale flight performance was tested using the 100% scale prototype and demonstrated that the competition aircraft will have desirable flight characteristics. With submaximal throttle, the prototype was able to take off easily and maneuver well, demonstrating how the competition aircraft with higher wing loading would have even better handling.

Figure 7.5: Full-sized Prototype Plane Flight in Flight (left) and on the ground (right).

7.3.2 Banner Tow Testing

Banner towing tests with the scaled prototype and other personal RC planes were performed to determine the resulting effects on aircraft performance and flight handling when adding a substantial drag force to the aircraft. The image to the right is of the scaled prototype in takeoff while towing a 9 ft long banner. Different deployment and tow points on the aircraft were experimented with to evaluate the effect of the moment induced by the banner's drag on stability and handling.

Figure 7.6: Banner Tow Testing on 40% Scale Prototype at Black Mountain

7.4 Risk Reduction and Preflight Checklist

Reducing the risk of a mid-flight failure or crash is of the utmost importance. Before flight, the aircraft conditions specified in Table 7.c will be verified and signatures will be required from the attending leads and advisors.

Table 7.c: List of safety checks performed prior to each test flight.

Preflight Checklist			
Propulsion		**Controls**	
Propeller Direction?	☐	Receiver Connection?	☐
Propeller Secured?	☐	Control Surfaces Responsive?	☐
Wiring Secured?	☐	Telemetry Software On?	☐
Motor Mounted Properly?	☐	Visual Inspection?	☐
Batteries Charged?	☐	**Aircraft**	
Battery Voltages +/- 3V?	☐	CG Verified?	☐
Receiver Pack Charged?	☐	Wing Box Secured?	☐
Receiver On?	☐	Hatches Closed?	☐
Connection Secured?	☐	Landing Gear Tight?	☐
Payload		**Signatures**	
Weight Verified?	☐	Pilot	_____
Passengers Restrained?	☐	Faculty Advisor	_____
Cargo Restrained?	☐	Project Manager	_____
Banner Secured?	☐	Date	_____

8. Performance Results

Data was collected from some of the testing and is presented below compared to the predicted or ideal results.

8.1 Subsystem Performance

8.1.1 Prototype Flight Test Results

The scaled prototype, with the same cubic wing loading as the competition plane at max weight, was flown and demonstrated that, with some practice, the competition plane would be controllable. Additionally, it was used to test banner towing; it towed a 9 ft long plastic tablecloth generic to dollar tree. In those tests, it showed that the selected banner tow point on the underside of the fuselage near the tail controls would be viable and not limit the flight performance of the aircraft.

Due to limited test flight availability, on account of weather conditions and other factors. The flight test data is limited but articulable through the Cooper–Harper ratings system which puts forward that from 1 to 10 how poorly a system functioned in terms of how much the pilot needed to compensate for poor performance. On the rating system described, the prototype aircraft scored a 3 in handling. The aircraft handled exceptionally well, needing some trim as standard to counteract the inherent pitch up and right roll that the aircraft exhibited in attempted straight and level flight occurring at around 50% throttle. The rating of 3 is mostly due to the high lift gained in those instances which had to be countered frequently, which is an issue that would be resolved in the competition aircraft with its increased wing loading. In turns and in other maneuvers where additional g-forces helped to mitigate the excess lift, the plane performed extremely well even being agile enough to recover from several instances of orientation loss due to poor conditions.

8.1.2 Ground Mission Timing Test Results

A GM prototype was utilized to simulate loading times in respect to each mission. From the GM prototype, finalized sub-components used for passenger and cargo loading were used to test M2 and M3 ground missions. For M2, the test featured the sub-components necessary to complete the respective GM for both loading and unloading the full length of the fuselage. For M3, the banner was pre-rolled to conserve time and measure the time it took to complete loading and unloading. The tabulated results are displayed in Table 8.1.

	M2		M3		Mech.	Total Best
	Loading	Unloading	Loading	Unloading	Hatches	
Avg. [s]	41.66	18.96	21.67	5.24	1.55	89.08
Avg. [s]	41.01	16.27	20.51	6.18	1.41	85.38

Table 8.1: Collected average time for GM configurations for M2 and M3

8.1.3 Structural Testing Results

The structural test on the wing spar verified the stiffness and reliability of the composite material to take aerodynamic and landing loads. As mentioned in §7.2.3, the test simulated a 6g maneuver load by introducing 90 lbs to the center of a simply supported configuration. Supports were added 13 inches from the center, which is the approximate centroid of the predicted lift distribution. The physical test yielded a relative displacement of 0.389 inches from the center to the support. The same ABAQUS FEA simulation from §5.2.1 was used to validate the findings from the physical test. Limitations of applying a spanwise distribution to the spar in the physical test may be the result of the discrepancy with respect to the simulation.

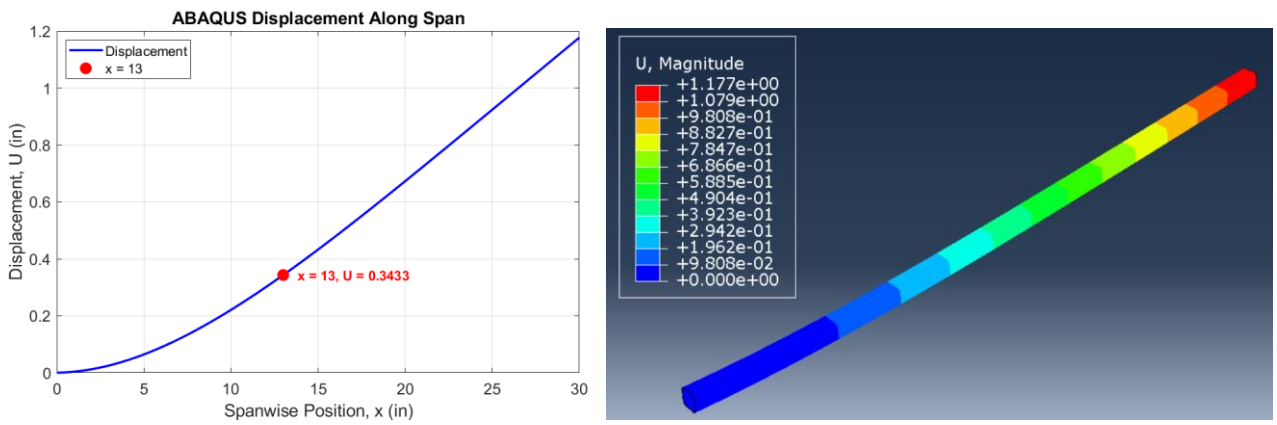


Figure 8.1: Validation of Physical Spar Test with FEA Simulation

9. Bibliography

[1] *30th Annual Design/Build/Fly Competition | 16–19 April 2026 | Wichita, KS (USA)*. AIAA; American Institute of Aeronautics and Astronautics. https://aiaa.org/dbf/

[2] *Previous Competitions*. (2026). AIAA - Shaping the Future of Aerospace; American Institute of Aeronautics and Astronautics. Retrieved February 19, 2026, from https://aiaa.org/dbf/previous-competitions/

[3] Raymer, D. (2012) Aircraft Design: A Conceptual Approach. 15th Edition. https://doi.org/10.2514/4.869112

[4] Lambe, Andrew & Martins, Joaquim. (2012). Extensions to the Design Structure Matrix for the Description of Multidisciplinary Design, Analysis, and Optimization Processes. Structural and Multidisciplinary Optimization. 10.1007/s00158-012-0763-y.

[5] *Parasite Drag Tool*. (2022). OpenVSP. https://openvsp.org/wiki/doku.php?id=parasitedrag

[6] *Beech Factory Airport*. (2026). National Weather Service. https://www.weather.gov/wrh/timeseries?site=KBEC

[7] *Bayesian game*. (2025). Wikipedia; Wikimedia Foundation. https://en.wikipedia.org/wiki/Bayesian_game#Elements

[8] *Choosing the Right Power for Your RC Airplane. (2016). Grayson Hobby.* https://graysonhobby.com/rc-airplane-power-guide

[9] Amandio, L., Marta, A., Afonso, F., Vale, J., Suleman, A., & Araujo, A. (2014). Stochastic optimization in aircraft design. *Engineering Optimization*, 267-272.

[10] Airfoil Tools. *Airfoiltools.com*. http://www.airfoiltools.com/. Accessed Feb. 17, 2025.

[11] Dulin, D. J. and Takahashi, T. T. (2015). *Design Implications of Elliptical Planform Wings*. *AIAA.*

[12] Nickel, K., Wohlfahrt, M. and Capt. Brown, E. (1994). *Tailless Aircraft in Theory and Practice.* AIAA.

[13] Sadraey, M. H. (2022). *Flight Stability and Control*. Synthesis Lectures on Mechanical Engineering. Springer.

[14] Yechout, T. R., Morris, S. L., Bossert, D. E., and Hallgren, W. F. (2003). *Introduction to Aircraft Flight Mechanics*. AIAA Education Series.

[15] Morr, G., & Rubin, D. (1978). *Drag investigations of towed target banners* (No. DRDMIT795).

[16] Morris-Thomas, M. T., & Steen, S. (2009). Experiments on the stability and drag of a flexible sheet under in-plane tension in uniform flow. *Journal of Fluids and Structures*, *25*(5), 815-830.

[17] Argentina, M., & Mahadevan, L. (2005). Fluid-flow-induced flutter of a flag. *Proceedings of the National Academy of Sciences*, *102*(6), 1829-1834.

[18] Fairthorne, R.A. (1930). *Drag of Flags.* Aeronautical Research Committee, Reports and Memoranda vol. 1345, pp. 887–891.

[19] Virot, E., Amandolese, X., & Hémon, P. (2013). Fluttering flags: an experimental study of fluid forces. *Journal of fluids and structures*, *43*, 385-401.

[20] *EP-0926-023 - Small Footprint 9V Battery Box with Battery Lead Clip*. (2026). ALLPARTS. https://www.allparts.com/products/ep-0926-023-small-footprint-9v-battery-box-with-battery-lead-clip?srsltid=AfmBOoqgOiZFwECKaVES5lZUHSpy7SVM3JHuKmw_3IxP85e6Jm2EymOH